The Fertility, Marriage, and Gender Equality Quandary

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Abstract

This paper documents that societies achieving any two of high fertility, dual parenthood, and gender income equality often fall short on the third. This shortfall is equally large in OECD and non-OECD countries, but the binding outcome differs: in the OECD, dual parenthood is the bottleneck; elsewhere, gender equality is the bottleneck. I develop a marriage-market model that links these outcomes through income sharing and joint fertility. Although conventional policies each leave at least one objective worse off, coordinating a female wage increase with an appropriately sized marriage benefit improves all three outcomes. Redistributing childcare from wives to husbands offers a complementary single-instrument route when both spouses gain and relative earnings converge. A calibration to Mexico finds scope for further redistribution as wages converge, but the range ends before equal sharing.

JEL classification: D13, J11, J12, J13, J16
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1 Introduction

Three demographic transformations have reshaped advanced economies over the past half-century. Fertility has fallen, often well below replacement. Stable two-parent families have become less common, and a rising share of children spend part of their childhood in single-parent households. Meanwhile, gender gaps in education and earnings have narrowed substantially and, in some settings, reversed (Goldin, 2014).

Each transformation has motivated a distinct policy agenda. Concerns about pension sustainability (Bongaarts, 2004) and long-run economic growth (Jones, 2022) have prompted pro-fertility measures ranging from baby bonuses to subsidized childcare. Interest in family stability, rooted partly in cultural considerations and partly in evidence on children's outcomes (Autor et al., 2019; Wasserman, 2020), has motivated pro-marriage policies such as joint taxation. Efforts to promote gender equality seek both to advance equal opportunity and to correct the misallocation of human capital, whose aggregate costs can be substantial (Hsieh et al., 2019). These agendas are often discussed separately, although the decisions they seek to influence are closely connected. Children change parents' market work, earning opportunities shape the attraction of marriage, and marriage changes the resources available for raising children. When does progress toward one objective support the others, and when does it come at their expense? I address this question in four steps.

First, I document how often the three outcomes coincide. I construct a global panel combining the total fertility rate from the World Development Indicators, children's co-residence with both parents from census microdata and the OECD Family Database, and the female share of labor income from the World Inequality Database. The sample covers 115 countries and 1,764 country-year observations. Only 2.8 percent of observations exceed the sample medians on all three dimensions, compared with 12.5 percent under statistical independence.

This aggregate shortfall conceals an important contrast. Dividing the sample into OECD and non-OECD groups and applying group-specific medians yields a shortfall of about 6 percentage points in each. An exact decomposition, however, reveals different bottlenecks: in each group, one outcome conflicts with both others, while those two are only weakly related. In the OECD, the bottleneck is dual parenthood. Its associations with fertility and with gender equality account for the shortfall, while the fertility–gender equality association is negligible. Outside the OECD, the bottleneck is gender equality. Its associations with fertility and with dual parenthood are the two largest contributions, while the fertility–dual parenthood association is the weakest. The challenge is therefore to understand both the rarity of joint attainment and why the binding outcome differs across country groups.

Second, I develop a tractable equilibrium marriage-market model that connects the three outcomes through income sharing and the time cost of children. Men and women draw idiosyncratic tastes for marriage. Marriage bundles an income transfer from husband to wife with shared fertil-

ity: the wife chooses fertility conditional on the transfer, and the husband evaluates the resulting consumption–fertility bundle when deciding whether to marry. The transfer adjusts to balance participation on the two sides of the market, making fertility and the terms of income sharing jointly determined.

The model links household choices to aggregate fertility, marriage, and the gender income gap. When mothers provide childcare, higher fertility reduces female market work. The income-sharing attraction of marriage, meanwhile, depends on the support available from husbands relative to women's own earning capacity. Together, these channels generate a policy tension. A larger marriage benefit or a lower time cost of children raises fertility and marriage but widens the gender income gap. Higher female wages narrow the gap but reduce fertility and marriage. The link between fertility and gender income inequality through maternal time use is well documented (Kleven et al., 2019). The marriage market adds an equilibrium channel: policy also changes the transfer that supports participation and finances marital fertility.

The gender wage gap helps organize this contrast. A large gap lowers the fertility level compatible with a given degree of income equality and strengthens the marital fertility premium, so fertility and marriage rise together while both draw mothers out of the market. Gender equality is then the hardest outcome to attain. Wage convergence relaxes that constraint but shrinks the income-sharing gains that support marriage, making dual parenthood harder to sustain.

Third, I examine whether policy can improve all three outcomes together. The three conventional instruments—a larger marriage benefit, higher female wages, and a lower childcare time cost—each leave at least one objective moving in the wrong direction. I show that these individual-instrument trade-offs, however, are not fundamental. Coordinating a female wage increase with an appropriately sized marriage-benefit adjustment raises fertility and marriage while narrowing the income gap. The marriage-benefit adjustment offsets the fertility decline induced by higher female wages without exhausting their income-equality gains, and the construction works at any interior baseline equilibrium regardless of the initial wage gap or preference parameters. Implementing such coordination requires calibrating the two instruments together: too little marriage-benefit support leaves fertility below its initial level, while too much reverses the income-gap improvement.

The two-instrument requirement raises a natural follow-up: can a single policy margin achieve the same joint improvement? I show that redistributing childcare obligations from wives to husbands, an increasingly popular lever among policymakers, holds such potential. Policies such as reserved paternity leave increasingly seek to change the division of care (Farré and González, 2019; Patnaik, 2019) because the maternal concentration of childcare may partly reflect persistent gender norms rather than current wages, technology, or preferences (Goldin, 2025; Deschamps et al., 2026; Albrecht et al., 2024; Xi and Zhou, 2026).

I establish three sufficient conditions under which redistribution raises fertility, marriage, and

gender income equality. First, it must lower the wife's effective childcare cost, accounting for the reduction in the husband's earnings available to finance her transfer. Second, women's earnings must improve relative to men's after fertility responds and creates additional care obligations. Third, the revised marital contract must leave room for the husband to gain after the transfer compensates the wife. Market clearing then shares the gains between spouses, raising both marriage premiums. The model also allows each unit of the father's time to produce fewer effective units of childcare. Lower paternal efficiency increases the earnings cost of his contribution and changes the relative-income response.

Furthermore, I characterize the limit of childcare redistribution. I establish an analytical ceiling strictly below full paternal provision. At or above that ceiling, the wife's effective-price condition and the husband's participation-gain condition cannot both hold. With equally efficient parents, the ceiling lies above equal sharing, leaving open how far redistribution can proceed in a particular economy. Locating the first condition failure requires following fertility and the equilibrium transfer as the childcare division changes, a task that I take up in the quantification.

Fourth, I quantify these mechanisms by calibrating the shared-childcare model to Mexico from 1990 to 2025.[1] Three time-varying inputs drive the exercise: gender-neutral productivity, the relative gender wage gap, and the division of childcare, with the last constructed from Mexico's National Survey on Time Use (ENUT). Four estimated parameters allow the model to track fertility and the broad movement in children's co-residence with both parents. A comparison with Mexico's female labor force participation rate from FRED, which is not targeted in estimation, also reproduces the rapid increase through 2008 and the subsequent deceleration.

The calibration supports the redistribution mechanism from 2001 onward: all three sufficient conditions hold at each year's allocation under equal parental efficiency from that date through 2025. In the early years of the sample, when the wage gap is largest, income convergence fails at the observed allocation. Following the equilibrium as fathers take on additional care, the paternal share at the first condition failure rises from approximately 24 percent in 2001 to 35 percent in 2025. This numerical boundary remains above the observed allocation throughout the years where the conditions hold, indicating scope for further redistribution, but below equal sharing. A complementary exercise holds the effective childcare division fixed and varies paternal efficiency. The lower efficiency boundary at which the conditions hold reaches approximately 0.81 by 2010, before rising after 2010 as the observed paternal share approaches the boundary of the sufficient-condition region. The calibrated transition therefore accommodates both a larger paternal contribution and a greater efficiency difference before the sufficient conditions fail.

Finally, a Shapley-value decomposition attributes the bulk of the fertility, family-structure, and

[1] Mexico offers a long time series, substantial wage changes during its demographic transition, and a trajectory broadly representative of transitions in many middle-income economies.

welfare changes to the narrowing relative wage gap. The increasingly equal division of childcare partially offsets the declines in fertility and dual parenthood. Consumption-equivalent measures show convergence toward married-male welfare among all other agent types, driven primarily by wage convergence. These results connect the local policy analysis to the longer transition: changing the division of care can buffer demographic declines even when broader wage changes continue to push in the opposite direction.

Appendix F explores a further connection across generations. Motivated by evidence that boys are more sensitive than girls to the absence of a biological father (Sommers, 2001; Autor et al., 2019; Wasserman, 2020), it allows lower current marriage to reduce the next generation's male relative human capital. Whether this feedback amplifies the demographic response or fades across generations remains a quantitative question for future research.

Related Literature

The paper connects four strands of research. The first links fertility to female labor supply through the time cost of children, beginning with Rosenzweig and Wolpin (1980) and continuing in quantitative models such as Erosa et al. (2016) and Blundell et al. (2016). Feyrer et al. (2008) document the reversal of the cross-country correlation between female labor force participation and fertility, suggesting that institutional adaptation can reconcile the two. Related work studies household production, childcare marketization, and parental leave (Bar et al., 2018; Greenwood, 2019; Kim and Yum, 2025); Doepke et al. (2023) and Greenwood et al. (2023) provide surveys. Doepke and Kindermann (2019) show that spousal disagreement over fertility is quantitatively important under limited commitment. I bring these household mechanisms into a marriage market, where the terms of income sharing adjust to policy and connect changes in mothers' constraints to fathers' willingness to marry.

In particular, the bilateral participation requirement result connects the analysis to the gender-revolution framework of Goldscheider et al. (2015), one of the most influential ideas in contemporary demography. They argue that women's entry into the labor market initially destabilizes families, while men's subsequent assumption of domestic responsibilities restabilizes them. The three sufficient conditions developed here formalize this insight and extend it in three directions: they identify the economic conditions under which restabilization occurs, they clarify why the same shift toward paternal involvement can succeed in one setting and fail in another, and they establish how far the reallocation of care can proceed before its benefits are exhausted. The key mechanism is the equilibrium transfer, which transmits a change in one spouse's time allocation to the other spouse's willingness to participate.

The second strand studies marriage-market clearing when men and women differ in productivity and preferences, following Becker (1973). Transferable-utility matching frameworks (Choo and

Siow, 2006; Chiappori, 2017) have been used to examine divorce law, household production, and technological change (Stevenson and Wolfers, 2007; Voena, 2015; Greenwood et al., 2016; Low, 2024). I abstract from within-gender wage heterogeneity to isolate the between-gender margin and follow Bisin and Tura (2019) in using an imperfectly transferable utility framework with micro-founded spousal utilities. In this setting, a change in the transfer changes fertility as well as consumption. The marital bundle and both spouses' valuations therefore depend on the equilibrium terms of income sharing.

The third strand examines how technological change reshapes the family (Greenwood et al., 2005a, 2016; Ngai and Petrongolo, 2017; Cao et al., 2024). Olivetti and Petrongolo (2016) and Feng et al. (2023) document long-run changes in gender gaps, while Hsieh et al. (2019) quantify the gains from improved talent allocation. I distinguish gender-neutral productivity from changes in relative wages and trace their effects on fertility, marriage, and income inequality through a common equilibrium.

The fourth strand studies how family structure shapes children's human capital. Boys appear particularly sensitive to disadvantage and the absence of a biological parent (Sommers, 2001; Bertrand and Pan, 2013; Chetty et al., 2016; Autor et al., 2019; Wasserman, 2020; Reeves, 2022; Autor et al., 2023; Frimmel et al., 2024). Kearney (2023) discusses the implications for inequality and mobility. Appendix F uses this evidence to motivate a feedback from current marriage to future male relative human capital.

The empirical analysis also draws on the descriptive benchmarking approach exemplified by the dartboard logic of Ellison and Glaeser (1997). I compare observed joint attainment with the frequency implied by independent outcomes having the same marginal attainment rates. An exact algebraic decomposition then separates the shortfall into pairwise contributions and a three-way residual. These tools establish the patterns that motivate the model, which supplies the economic structure needed to study policy responses.

The remainder of the paper follows this progression. Section 2 documents the empirical shortfall and its composition. Section 3 develops the baseline model and compares individual and coordinated policy instruments. Section 4 establishes the conditions for childcare redistribution to improve all three outcomes and derives an analytical ceiling on those conditions. Section 5 evaluates the mechanisms and redistribution boundaries in Mexico. Section 6 concludes. Appendix F develops the cross-generational extension.

2 Motivating Facts

The empirical analysis asks two questions: how often do high fertility, widespread dual parenthood, and gender income equality coincide, and which pairwise associations account for their limited

overlap? I first construct a global panel and compare joint attainment with an independence benchmark. I then examine OECD and non-OECD subsamples separately and decompose their shortfalls.

2.1 Data Sources and Construction

Data sources

Three categories of data, drawn from four source datasets, are combined to construct the analysis sample:

(i) **Total fertility rate** from the World Development Indicators (WDI), measured as children per woman.

(ii) **Dual parenthood** from two complementary sources. The IPUMS International (IPUMSI) database provides census microdata on children's household composition, from which I compute the share of children living with both parents. The OECD Family Database provides analogous national-level statistics on children's living arrangements. For country-years with data from both sources, I average the two measures; otherwise I use whichever is available.

(iii) **Gender income equality**, measured as the female share of national labor income from the World Inequality Database (WID) (Chancel et al., 2022). This measure is more comprehensive than the earnings gap among workers, since it incorporates differences in labor-force participation.[2]

Panel construction

I build a master country-year panel by taking the union of all country-years appearing in any of the four source datasets, then merge the variables by country and year. Missing values between observed data points within a country are filled by linear interpolation. The analysis retains only complete cases: country-years for which all three variables (fertility, dual parenthood, and female income share) are observed or interpolated.

Sample overview

The resulting *full sample* contains 1,764 country-year observations across 115 countries. Data availability for the dual-parenthood measure naturally divides the sample into two groups. Countries for which at least one year of OECD Family Database data exists form the *OECD subsample*

[2]Replacing the WID female income share with the OECD unadjusted gender wage gap among employed workers preserves the OECD pattern (Appendix A.3).

(879 observations, 39 countries). The remaining countries, whose dual-parenthood data come exclusively from IPUMSI census microdata, form the *non-OECD subsample* (885 observations, 76 countries). Table 1 summarizes the three samples.

Sample	Countries	Obs	Dual parenthood source	Gender equality measure
Full	115	1,764	IPUMSI + OECD	Female income share (WID)
OECD	39	879	OECD Family Database (+ IPUMSI)	Female income share (WID)
Non-OECD	76	885	IPUMSI only	Female income share (WID)

Table 1. Summary of Analysis Samples

Notes: All samples use the total fertility rate from the WDI for fertility and the female labor income share from the WID for gender equality. The OECD subsample includes countries with at least one year of OECD Family Database dual-parenthood data. Where both IPUMSI and OECD data are available for a country-year, the two are averaged. Missing values are interpolated linearly within country between observed data points.

The three measures capture distinct aspects of the demographic transition. The total fertility rate summarizes fertility, while children's co-residence with both parents measures the family environment directly. The latter avoids relying on marriage or nonmarital-birth statistics, which need not correspond closely to children's living arrangements. The female labor income share incorporates both earnings opportunities and participation, allowing the analysis to capture gender differences beyond wages among employed workers.

2.2 The Trinity Shortfall

Variable construction

Within each sample, I classify observations relative to the sample-specific median of each outcome. "High fertility" (F) denotes a total fertility rate above its median, "dual parenthood" (M) denotes a share of children living with both parents above its median, and "gender equality" (G) denotes a female income share above its median. These definitions measure relative attainment within a sample and yield approximately balanced categories. All classifications use strict inequalities, so observations exactly at a cutoff remain outside the corresponding attainment group.

Table 2 reports the cutoffs for all three samples. Their variation highlights the heterogeneity that motivates the subsample analysis. The median TFR ranges from 1.56 in the OECD to 3.24 in non-OECD countries. The dual-parenthood median ranges from 70.5 percent in non-OECD countries to 83.7 percent in the OECD. The female income share median ranges from 29.4 percent in non-OECD to 37.4 percent in OECD countries.

The benchmark asks how often all three indicators would equal one if their observed marginal attainment rates were combined independently. Writing p_F, p_M, and p_G for those rates, expected joint attainment is $p_{111}^{\text{ind}} = p_F p_M p_G$. Independent indicators with marginal attainment probabilities

Sample	High fertility (F)	Dual parenthood (M)	Gender equality (G)
Full	TFR > 2.00	Both parents $> 80.1\%$	Female share $> 34.1\%$
OECD	TFR > 1.56	Both parents $> 83.7\%$	Female share $> 37.4\%$
Non-OECD	TFR > 3.24	Both parents $> 70.5\%$	Female share $> 29.4\%$

Table 2. Median Cutoffs for Binary Indicators

Notes: Displayed cutoffs are rounded; calculations use the numerical thresholds returned by the statistical routine. TFR denotes the total fertility rate, children per woman. "Both parents" is the share of children living with both parents. "Female share" is the female share of national labor income. Appendix Table A.1 reports the cutoffs alongside additional sample statistics.

of one-half yield 12.5 percent.

Full-sample results

Figure 1 presents the joint distribution in the full sample. Only 2.8 percent of country-year observations achieve the trinity F+M+G, compared with the 12.5 percent independence benchmark. The largest categories are "F only" (26.9 percent) and "M+G" (25.7 percent). Thus, observations with above-median fertility are concentrated below the cutoffs for the other two outcomes, while those above both the dual-parenthood and gender-equality cutoffs tend to have below-median fertility.

Country-clustered bootstrap inference (1,000 replications) yields a 95 percent percentile interval of $[-11.87, -6.48]$ percentage points for the shortfall, with a normal-approximation p-value below 0.001. The trinity is significantly rarer than independence would predict.

Motivation for subsample analysis

The full sample pools countries with total fertility rates ranging from below 1 to nearly 7, dual-parenthood shares from 26 to 96 percent, and female income shares from 3 to 50 percent. Median splits within such a heterogeneous pool may conflate differences in levels with genuine trade-offs within comparable groups of countries. To address this concern, I split the sample into OECD and non-OECD countries, re-estimating all medians and binary indicators within each subsample.

2.3 OECD and Non-OECD Results

Figure 2 presents Venn diagrams for the two subsamples. Table 3 reports the trinity shortfall for all three samples.

Two findings stand out. First, the shortfall is present and statistically significant in both subsamples, despite the very different country compositions and cutoff levels. Second, the magnitude

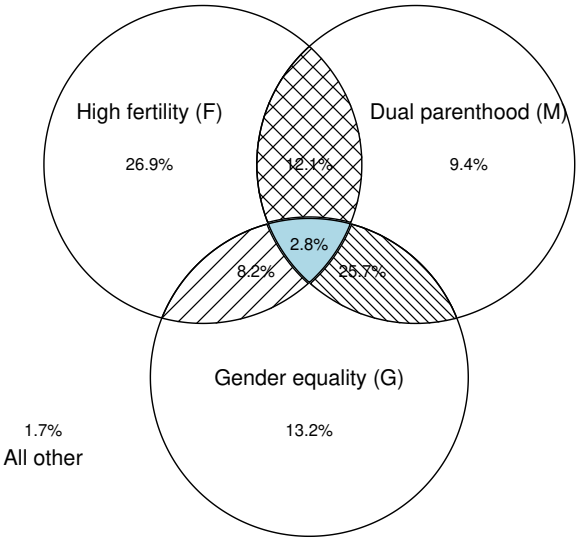
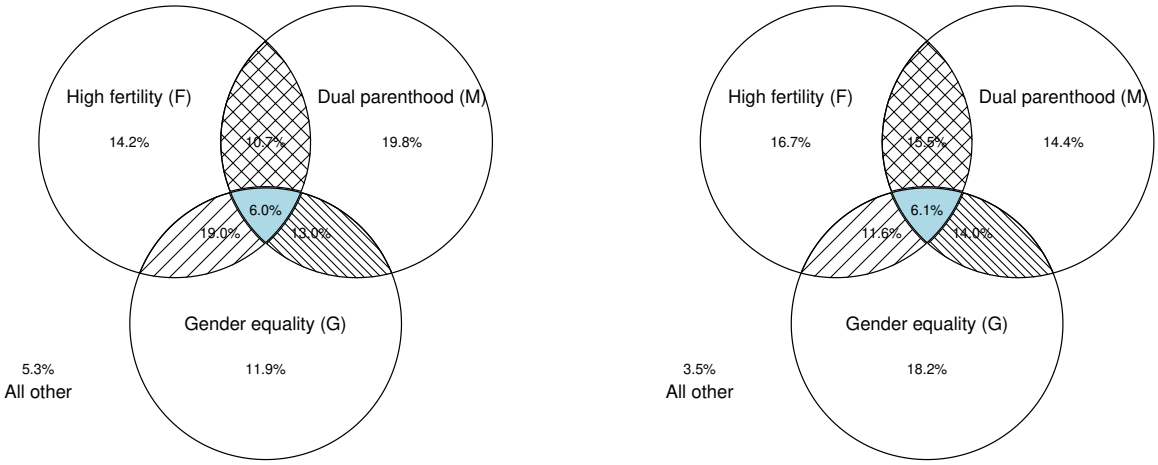


Figure 1. Three-Way Trade-Off in the Full Sample ($N = 1{,}764$)

Notes: Venn diagram showing the joint distribution of high fertility (F), dual parenthood (M), and gender equality (G), each defined at the full-sample median. Under statistical independence, the central overlap (F+M+G) would contain 12.5 percent of observations. The empirical independence benchmark is 12.49 percent.



(a) OECD ($N = 879$, 39 countries)

(b) Non-OECD ($N = 885$, 76 countries)

Figure 2. Three-Way Trade-Off: OECD and Non-OECD Subsamples

Notes: Each subsample uses its own median cutoffs. Under statistical independence, the central overlap would contain approximately 12.5 percent. See Table 3 for bootstrap inference.

Sample	Observed (%)	Independence (%)	Shortfall (pp)	Ratio	95% CI (pp)	p-value
Full	2.83	12.49	-9.65	0.23	$[-11.87, -6.48]$	< 0.001
OECD	6.03	12.37	-6.34	0.49	$[-10.34, -2.19]$	0.002
Non-OECD	6.10	12.49	-6.38	0.49	$[-9.85, -0.98]$	0.005

Table 3. Trinity Shortfall Across Samples

Notes: Shortfall is observed minus independence benchmark joint-attainment share. Ratio is observed divided by independence benchmark. Bootstrap inference uses 1,000 replications clustered by country. Percentile confidence intervals are reported. The p-value is the two-sided normal approximation based on the observed coefficient and bootstrap standard error.

of the shortfall is remarkably similar: 6.34 percentage points in the OECD and 6.38 percentage points in non-OECD countries, with observed-to-independence ratios of 0.49 in both cases.[3]

While the levels of the shortfall coincide, the Venn diagrams already suggest which outcome is missing in each group. In the OECD, the largest single category is "M only" (19.8 percent), alongside a sizable "F+G" category (19.0 percent) and "F only" (14.2 percent): high fertility and gender equality frequently coincide, but observations attaining both typically lack dual parenthood. In non-OECD countries, "G only" (18.2 percent), "F only" (16.7 percent), and "F+M" (15.5 percent) are all prominent: fertility and dual parenthood coincide more readily, while gender equality tends to occur without them. The next subsection quantifies these patterns.

2.4 Decomposing the Shortfall

Decomposition method

To identify the associations underlying the shortfall, I use an exact identity for three binary variables:

$$p_{111} - p_F p_M p_G = \underbrace{\delta_{FM} \cdot p_G}_{\text{F-M contribution}} + \underbrace{\delta_{FG} \cdot p_M}_{\text{F-G contribution}} + \underbrace{\delta_{MG} \cdot p_F}_{\text{M-G contribution}} + \varepsilon, \tag{1}$$

where $\delta_{XY} = P(X{=}1, Y{=}1) - P(X{=}1)P(Y{=}1)$ measures the excess joint attainment of a pair relative to independence, and ε captures the remaining three-way interaction. Each pairwise contribution is weighted by the marginal attainment probability of the third outcome. A negative δ_{XY} means that the pair co-occurs less often than independence would imply and therefore contributes negatively to the trinity shortfall.

The identity provides a direct accounting of the shortfall in probability units. Appendix A.2 supplements it with mutual information statistics, which measure dependence through the reduction in uncertainty about one variable from observing another. These statistics provide complementary

[3]The full-sample shortfall (9.65 percentage points) is larger than either subsample's, reflecting additional compositional dependence created by pooling countries at very different levels of development.

diagnostics, while the pairwise decomposition identifies the associations that contribute most to the shortfall studied here.

Different bottlenecks in different settings

Table 4 presents the decomposition for the OECD and non-OECD subsamples.

	OECD		Non-OECD	
Component	Contribution (pp)	% of shortfall	Contribution (pp)	% of shortfall
F-M tension	−4.02	63.4%	−1.71	26.7%
F-G tension	+0.07	−1.1%	−3.57	55.9%
M-G tension	−2.86	45.1%	−2.42	37.9%
Three-way residual	+0.47	−7.4%	+1.31	−20.5%
Total shortfall	−6.34	100%	−6.38	100%

Table 4. Pairwise Contribution Decomposition of the Trinity Shortfall

Notes: The identity in (1) decomposes the total shortfall into contributions from each pairwise tension plus a three-way residual. A negative contribution means the corresponding pair of outcomes repels each other, making the trinity rarer. A positive contribution means the pair tends to co-occur, partially offsetting the shortfall. Bootstrap inference is reported in Appendix Table A.4.

In OECD countries, the fertility–dual parenthood contribution accounts for 63.4 percent of the shortfall, and the dual parenthood–gender equality contribution for a further 45.1 percent. By contrast, the fertility–gender equality contribution is negligible (−1.1 percent): the corresponding median-split indicators are approximately uncorrelated. Dual parenthood is therefore the bottleneck.[4] High fertility and gender equality are compatible with each other, but each tends to occur without dual parenthood.

In non-OECD countries, the pattern centers on gender equality instead. The fertility–gender equality contribution accounts for 55.9 percent of the shortfall and the dual parenthood–gender equality contribution for 37.9 percent, while the fertility–dual parenthood contribution is the smallest, at 26.7 percent. Gender equality is therefore the bottleneck: it is the outcome that tends to be missing when fertility and dual parenthood are attained. All three pairwise contributions are negative, while the positive three-way residual partially offsets them. The positive residuals in both samples also explain why the negative pairwise contributions can sum to more than 100 percent of the shortfall.

Bootstrap confidence intervals (Appendix Table A.4) support this reading more strongly in the OECD than outside it. In the OECD, both contributions involving dual parenthood are significantly

[4] I use "bottleneck" in a descriptive sense: the outcome shared by the two largest negative pairwise contributions, whose complementary pair contributes least. The term identifies which outcome is hardest to combine with the other two. It does not assign a causal interpretation to the associations or indicate which outcome policy should target.

negative (F-M, $p = 0.003$; M-G, $p = 0.044$), while the F-G contribution is indistinguishable from zero ($p = 0.964$). In non-OECD countries, the F-G contribution is significantly negative ($p = 0.005$). The M-G contribution has the same sign but is less precisely estimated ($p = 0.105$), and the F-M contribution is not significant ($p = 0.253$). The non-OECD bottleneck is thus anchored by the fertility–gender equality tension, while the ranking of the other two contributions rests on point estimates.

The OECD pattern also persists with an alternative gender-equality measure. Replacing the WID female income share with the OECD unadjusted gender wage gap among employed workers reduces the direct role of participation in the measure. The fertility–dual parenthood contribution then accounts for virtually the entire shortfall, while the fertility–gender equality contribution remains negligible (Appendix A.3). The dual parenthood–gender equality contribution becomes small, suggesting that this conflict operates largely through participation. The central feature of the OECD pattern survives: fertility and gender equality remain unrelated, and the shortfall remains concentrated in a pair involving dual parenthood.

2.5 Summary

Two findings motivate the theory. First, joint attainment of high fertility, widespread dual parenthood, and gender income equality is substantially less common than independence would imply across 115 countries. The shortfall remains statistically significant when OECD and non-OECD countries are considered separately. Second, the similar magnitudes of the two subsample shortfalls conceal different bottlenecks: dual parenthood in the OECD, where fertility and gender equality are approximately unrelated, and gender equality outside it, where the fertility–dual parenthood tension is the weakest of the three.

These conclusions concern relative attainment under the median-split definitions. In particular, the OECD bottleneck rests on the negligible fertility–gender equality contribution for the binary indicators. In the original variables, the OECD correlation between fertility and the female income share is -0.42, the strongest of the three (Appendix A.1). The bottleneck therefore describes joint attainment above group medians rather than linear association across the full distribution. The decomposition identifies which outcome limits joint attainment without assigning a causal interpretation to the associations.

The next step is to examine the mechanisms that can connect the three outcomes and the policy trade-offs those mechanisms create. This requires a framework in which fertility, marriage, and labor income are jointly determined, with the terms of the marital contract adjusting to changes in the economic environment. The following section develops that framework.

3 Model

This section develops an equilibrium marriage-market model linking fertility, marriage, and the gender income gap. The baseline assigns childcare to mothers to isolate two mechanisms: children reduce women's labor supply, and marriage transfers resources that support fertility. The equilibrium transfer connects household choices to participation on both sides of the marriage market. After characterizing this equilibrium, I use it to interpret the empirical patterns and compare policy instruments.

3.1 Setup

Individuals are indexed by gender $g \in \{♂, ♀\}$, with each gender comprising half the population. Utility depends on consumption c and fertility n through a constant-elasticity-of-substitution aggregator:

$$u(c,n) = \left[(1-\beta)c^{\frac{\rho-1}{\rho}} + \beta n^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}, \qquad \beta \in (0,1), \quad \rho > 1. \tag{2}$$

Following Jones and Schoonbroodt (2010) and Carlos Córdoba and Ripoll (2019), I assume that consumption and fertility are substitutes. This assumption allows rising opportunity costs to reduce fertility as wages grow. Utility in (2) is also well-defined for childless individuals, with $u(c,0) = (1-\beta)^{\rho/(\rho-1)}c$.

To focus on between-gender differences, wages are uniform within each gender and strictly positive.[5] Let $w^{♂}$ and $w^{♀}$ denote male and female wages, and define the gender wage gap as $\Gamma^w = w^{♂}/w^{♀}$. A higher ratio indicates a larger male wage advantage. Wages are exogenous in the static model.[6]

Single individuals

Single men supply one unit of labor, consume their earnings, and remain childless:

$$V^{♂,s} = u(w^{♂}, 0).$$

[5] Abstracting from within-gender wage and education heterogeneity shuts down assortative matching, which would require additional structure (Chiappori, 2017). Figure A.3 reports a spousal earnings gap among newly-wed, childless U.S. couples of 20 to 30 percent, compared with approximately 10 percent among young workers, motivating attention to within-couple gender differences.

[6] In life-cycle models such as Erosa et al. (2016) and Blundell et al. (2016), labor supply also affects subsequent human capital and wages. The static model isolates the contemporaneous labor-supply and marriage-market channels.

Single women can have children without support from the father.[7] They choose consumption, market work, and fertility to solve

$$V^{♀,s} = \max_{c^{♀,s},l^s,n^s} \; u(c^{♀,s}, n^s)$$

subject to

$$c^{♀,s} = w^{♀} l^s, \qquad l^s = 1 - \chi n^s, \qquad c^{♀,s} \geq 0, \quad l^s \geq 0, \quad n^s \geq 0.$$

Each child requires $\chi > 0$ units of the mother's time. A single mother therefore faces a direct choice between market consumption and children: more children leave less time to earn the income that finances consumption. Following De La Croix and Doepke (2003), fertility is continuous.

The setup captures the competition between consumption and fertility for household resources. Purchased childcare (Bar et al., 2018) and explicit investments in child quality (Becker and Lewis, 1973) are left for extensions. Section 4.5 discusses how outsourced care would differ from the time-cost changes studied here.

Married individuals

Marriage changes the resources available for raising children. In the baseline, husbands provide no childcare, supply one unit of labor, and transfer a share α of their earnings to their wives. The transfer captures the income-sharing function of marriage. Individuals take it as given, while its equilibrium value equates the number of men and women willing to marry.

The necessary resource restrictions are

$$-\frac{w^{♀}}{w^{♂}} < \alpha < 1.$$

The lower bound ensures positive resources for the wife before childcare, and the upper bound leaves positive consumption for the husband. Negative transfers are permitted in principle, although the interior equilibrium characterized below has $\alpha > 0$.

Fertility decision rights are assigned to the wife. She chooses fertility conditional on the committed transfer, and the husband evaluates the resulting bundle when deciding whether to

[7] The model abstracts from child support. Introducing enforceable support obligations would change both the resources available to single mothers and men's outside option, and would therefore require a corresponding change in the marriage-market analysis. According to the analysis by the Annie E. Casey Foundation using 2020-22 Current Population Survey data, only 23 percent of U.S. female-headed families with children under 18 receive any child support in the prior year.

marry.[8] His marital value is

$$V^{\male,m} = \lambda u\big((1-\alpha)w^{\male}, n^m\big), \qquad \lambda > 0.$$

The wife solves

$$V^{\female,m} = \max_{c^{\female,m}, l^m, n^m} \lambda u(c^{\female,m}, n^m)$$

subject to

$$c^{\female,m} = \alpha w^{\male} + w^{\female} l^m, \qquad l^m = 1 - \chi n^m, \qquad c^{\female,m} \geq 0, \quad l^m \geq 0, \quad n^m \geq 0.$$

Parameter λ is a utility shifter that scales the utility from marriage, reflecting factors such as social norms. In the baseline model, I assume that λ is gender-neutral.[9]

The within-marriage transfer creates an income effect for the wife: at a given wage, more support allows her to choose both more consumption and more children. A higher female wage raises income while also making time spent caring for children more expensive. With $\rho > 1$, the opportunity-cost effect dominates and desired fertility falls at a fixed nonnegative transfer. This is the substitution mechanism emphasized by Greenwood et al. (2005a). A higher male wage raises the resources available for child-rearing at a fixed positive transfer, consistent with the income channel in Jakobsen et al. (2022). The equilibrium analysis adds a further step by allowing the transfer itself to respond to these wage changes.

Marriage market

At the start of the period, each individual draws independent, multiplicative taste shocks for marriage and singlehood. The shocks have a common Fréchet distribution with shape parameter $\theta > 0$. Individuals choose the alternative with the higher taste-adjusted value, yielding

$$\mathcal{M}^g = \frac{(V^{g,m})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta}, \qquad g \in \{\male, \female\}. \tag{3}$$

A higher marriage-to-singlehood utility ratio attracts more participants; a larger θ makes participation more responsive to that ratio. Appendix D.1 derives (3) directly from the taste-shock distribution.

[8] The timing commits the transfer before fertility is chosen and assigns the fertility decision to the wife. Introducing subsequent bargaining, consent, or renegotiation after children are born would require a different contracting environment.

[9] With gender-specific benefits, the ratio $\lambda^{\female}/\lambda^{\male}$ no longer cancels from (4), so the equilibrium transfer depends on it. A common proportional change in both benefits—the experiment in Propositions 1 and 4—still leaves α^* and household fertility unchanged. Separating the two benefits would require direct data on within-marriage transfers.

With equal masses of men and women, market clearing requires $\mathcal{M}^{\male} = \mathcal{M}^{\female} = \mathcal{M}$. Because the two genders have the same taste-shock distribution, this is equivalent to

$$\frac{V^{\male,m}}{V^{\male,s}} = \frac{V^{\female,m}}{V^{\female,s}}. \tag{4}$$

Equation (4) determines the equilibrium terms of marriage. The transfer must make marriage equally attractive relative to singlehood on both sides of the market, taking account of the fertility response it induces. The pair (α, n^m) therefore describes the marital bundle: the transfer clears the market, and the wife's household problem determines fertility conditional on that transfer.

The competitive interpretation follows Choo and Siow (2006): individuals are atomistic and take the terms of marriage as given. In the present setting, those terms also affect the bundle being exchanged. A larger transfer changes both the division of consumption and the number of children, so both spouses' valuations respond to the wife's fertility choice.

Aggregate outcomes

The three outcomes of interest are aggregate fertility, marriage, and the gender income gap. Their connection through female labor supply is

$$n = \mathcal{M}n^m + (1-\mathcal{M})n^s, \qquad l^{\female} = 1 - \chi n, \qquad \Gamma^y = \frac{y^{\male}}{y^{\female}} = \frac{\Gamma^w}{l^{\female}}. \tag{5}$$

Fertility is measured per woman. Labor incomes are measured before private transfers: $y^{\male} = w^{\male}$ and $y^{\female} = w^{\female} l^{\female}$.

The model measures family structure through adult marriage, while the empirical analysis measures children's co-residence with both parents. The mapping between them depends on the marriage rate and relative marital fertility, $q = n^m/n^s$. The child-based measure increases in both objects. It therefore moves with marriage in all individual-policy comparisons and in Proposition 5, where $\mathcal{M}$ and q move in the same direction or q remains unchanged.[10]

The last relation captures the distinction between wages and incomes. Even at a fixed wage gap, the income gap widens when women spend more time outside the market caring for children. Fertility therefore affects gender income inequality through labor supply, while marriage affects fertility by changing household resources and the composition of families.

[10] With relative marital fertility $q = n^m/n^s$, the child-based share equals $\frac{\mathcal{M}q}{1+\mathcal{M}(q-1)}$. Their common comparative-static direction permits a shared interpretation in these experiments, although their levels differ and other changes may alter fertility composition. Appendix D.4 provides the mapping.

3.2 Equilibrium Characterization

Before solving the model, I state the equilibrium formally.

Definition 1 (Competitive equilibrium)**.** *Given primitives* $(\beta, \rho, \chi, \lambda, \theta)$ *and wages* $(w^{♂}, w^{♀})$*, a competitive equilibrium consists of single women's choices, married women's choices, a transfer share* α*, and marriage rates satisfying:*

1. *Single women optimize subject to their resource and time constraints. Single men remain childless and consume their earnings.*
2. *Wives optimize conditional on the transfer and wages.*
3. *Husbands evaluate marriage at the fertility chosen by wives.*
4. *Taste-adjusted choices generate the participation rates in* (3)*.*
5. *The transfer clears the marriage market as in* (4)*.*

The equilibrium can be solved in closed form. To keep the expressions manageable, define

$$\begin{aligned} r &\equiv \frac{\rho - 1}{\rho}, \\ C &\equiv \left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho} (w^{♀})^{\rho-1} + \chi, \\ K &\equiv 1 - \frac{\beta}{(1-\beta)(Cw^{♂})^{r}}. \end{aligned} \tag{6}$$

The wife's first-order condition then gives

$$n^{s} = \frac{1}{C}, \qquad n^{m} = \frac{1 + \alpha\Gamma^{w}}{C}. \tag{7}$$

Equation (7) makes the income-sharing mechanism transparent. Single and married women face the same monetary price of a child, $w^{♀}\chi$. Marriage adds the transfer while leaving that price unchanged. Marital fertility is consequently single fertility multiplied by the increase in resources relative to the wife's own earning capacity. If the transfer were zero, the two fertilities would coincide.

The closed-form solution must also respect the household time constraints. I therefore impose the following restrictions.

Assumption 1 (Interior baseline equilibrium)**.** *The primitives satisfy*

$$Cw^{♂} > \left(\frac{\beta}{1-\beta}\right)^{1/r}, \qquad \frac{\chi(1+\Gamma^{w})}{C(1+K^{1/r}\Gamma^{w})} < 1. \tag{8}$$

Policy comparisons remain within the region with strictly positive consumption and market labor supply.

The first inequality implies $0 < K < 1$ and delivers strictly positive consumption after the transfer. The second ensures that the associated marital fertility leaves the wife strictly positive time for market work. Together, they ensure that the marriage-market solution corresponds to a feasible interior household allocation. Single women's labor supply is already positive because $C > \chi$.

Lemma 1 (Baseline equilibrium)**.** *Under Assumption 1, the interior equilibrium is unique and satisfies*

$$\alpha^* = \frac{1 - K^{1/r}}{1 + K^{1/r}\Gamma^w} \in (0, 1), \qquad n^m = \frac{1 + \Gamma^w}{C(1 + K^{1/r}\Gamma^w)}. \tag{9}$$

The transfer and household fertilities are independent of λ and θ.

The proof appears in Appendix D.3. The independence from λ has a useful interpretation. A common proportional increase in the benefit of marriage makes marriage more attractive to both genders. It raises participation while preserving the transfer needed to balance the two sides, because the common benefit cancels from (4). Similarly, θ governs participation's responsiveness to gains, while the equality of those gains determines the equilibrium transfer.

Lemma 2 (Marriage gains)**.** *In the baseline equilibrium,*

$$\begin{aligned} \frac{V^{♀,m}}{V^{♀,s}} &= \lambda(1 + \alpha^*\Gamma^w), \qquad \frac{n^m}{n^s} = 1 + \alpha^*\Gamma^w > 1, \\ \mathcal{M} &= \frac{[\lambda(1 + \alpha^*\Gamma^w)]^\theta}{1 + [\lambda(1 + \alpha^*\Gamma^w)]^\theta}. \end{aligned} \tag{10}$$

Lemma 2 states the economic logic compactly. Women's gains from marriage have a common marriage-benefit component, λ, and an income-sharing component, $1 + \alpha^*\Gamma^w$. The latter depends on what the transfer is worth relative to the wife's own earning capacity. Because the price of fertility is unchanged, the same factor scales consumption, fertility, and utility across being single versus being married. Market clearing equates the spouses' marriage-to-singlehood utility ratios, while their utility levels can differ. Appendix D.2 provides the proof.[11]

[11] For completeness, Appendix D.1 also writes the husband's participation condition as a transfer schedule conditional on marriage and fertility. That schedule shows how much he is willing to transfer for a given marital bundle. It complements the wife's fertility first-order condition: she chooses fertility, and he decides whether the resulting contract is worth entering.

3.3 The Three-Way Trade-Off

Equations (5) and (10) connect the three outcomes. Income sharing makes marriage attractive by adding to the wife's resources, while the children it supports reduce her market work. A larger male wage advantage can strengthen the income-sharing attraction of marriage, but it also widens the income gap directly. Fertility then compounds that gap through female labor supply.

These links constrain the joint attainment of ambitious targets. High fertility requires maternal time, making income equality harder to sustain at a given wage gap. Narrowing the wage gap eases that inequality but reduces the value of a husband's transfer relative to the wife's own earning capacity. At a fixed common marriage benefit, maintaining participation therefore becomes more demanding as the wage gap narrows.

The transfer bound makes this dependence on the economic environment explicit. Every interior baseline equilibrium satisfies

$$\frac{1}{\lambda}\left(\frac{\mathcal{M}}{1-\mathcal{M}}\right)^{1/\theta} < 1 + \Gamma^{y}(1-\chi n). \tag{11}$$

At fixed λ and χ, sufficiently ambitious joint targets violate (11). Raising the marriage target increases the left side, while raising fertility or lowering the income-gap target reduces the right side. Appendix D.9 derives this necessary restriction.

Rationalizing the cross-country patterns

Before turning to policy, I examine how the wage gap changes the links among outcomes. Section 2.4 identifies gender equality as the bottleneck outside the OECD and dual parenthood as the bottleneck within it. The model highlights two mechanisms relevant to this contrast: the fertility level compatible with an income-equality target, and the marital fertility premium, which governs both the composition channel from marriage to fertility and the income-sharing gains that support marriage.

Lemma 3 (The wage gap and the bottleneck)**.** *Consider two economies sharing primitives* $(\beta, \rho, \chi, \lambda, \theta)$ *and male wage* $w^{\male}$*, with* $w^{\male} > w^{\female} > 0$ *and interior equilibria, but differing in the gender wage gap* $\Gamma^{w} = w^{\male}/w^{\female}$*.*

(i) ***F–G amplification.*** *The gender income gap satisfies* $\Gamma^{y} = \Gamma^{w}/(1-\chi n)$*. The maximum fertility consistent with a gender-equality threshold* $\bar{\Gamma} > \Gamma^{w}$ *is*

$$\bar{n}(\Gamma^{w}) = \frac{1}{\chi}\left(1 - \frac{\Gamma^{w}}{\bar{\Gamma}}\right),$$

which is strictly decreasing in Γ^{w} *and approaches zero when* $\Gamma^{w} = \bar{\Gamma}$*.*

(ii) ***F–M coupling.*** *The marital fertility premium* $q - 1 = (1 - g)\Gamma^w/(1 + g\Gamma^w)$*, where* $g = K^{1/r} \in (0, 1)$*, is strictly increasing in* Γ^w *through two reinforcing channels: the direct effect of a larger wage gap and the indirect effect through a lower* g*. The composition elasticity of aggregate fertility with respect to marriage,* $\varepsilon_{n,\mathcal{M}} = \mathcal{M}(q - 1)/[1 + \mathcal{M}(q - 1)]$*, is also increasing in* Γ^w*.*

The proof appears in Appendix D.5. At a large wage gap, the two parts combine to make gender equality the bottleneck. Part (i) tightens the fertility level compatible with an income-equality target. Part (ii) strengthens the composition channel, so fertility and marriage tend to move together. Indeed, the marriage benefit, the childcare time cost, and female wages all move n and $\mathcal{M}$ in the same direction (Propositions 1–3 below). Fertility and marriage thus reinforce each other while each works against income equality. This matches the non-OECD pattern in Table 4, where the two contributions involving gender equality are the largest (55.9 and 37.9 percent) and the fertility–dual parenthood contribution is the smallest (26.7 percent).

As the wage gap narrows, the same mechanisms shift the bottleneck. The fertility ceiling compatible with income equality rises, so fertility and gender equality can coexist. At the same time, the marital fertility premium $q - 1 = \alpha^*\Gamma^w$ declines. By Lemma 2, this is also the income-sharing component of women's gains from marriage: the husband's transfer adds less relative to the wife's own earning capacity, and at a given common benefit λ fewer women find marriage worthwhile. As the feasibility bound (11) indicates, sustaining marriage then depends increasingly on the common benefit rather than on income sharing. Dual parenthood becomes the outcome that is hardest to combine with the other two. This matches the OECD pattern, where the contributions involving dual parenthood dominate (63.4 and 45.1 percent) and the fertility–gender equality contribution is negligible (-1.1 percent).[12]

The marriage–gender equality channel operates at both wage-gap levels. Holding household fertilities fixed, more marriage raises aggregate fertility through composition, reduces female market work, and widens the income gap. This is consistent with the negative M-G contributions in both subsamples (45.1 and 37.9 percent). Because both bottlenecks share this pair, the contrast between them comes down to which outcome fertility conflicts with: gender equality at large wage gaps, dual parenthood at narrow ones.

The lemma characterizes structural links rather than the cross-country associations themselves. In the baseline model, fertility and marriage move in the same direction. The model therefore explains why the fertility–gender equality constraint relaxes and why marriage loses its income-sharing support as wages converge, but it does not by itself generate the negative fertility–dual

[12] Doepke et al. (2023) emphasize the growing compatibility of fertility and gender equality in advanced economies and the institutional adaptations that support it. The model isolates a complementary wage channel: holding childcare requirements and the equality target fixed, a narrower wage gap raises the fertility ceiling compatible with that target.

parenthood association observed in the OECD. Once the composition channel weakens, that association must come from primitives that move fertility and marriage separately, which lie outside the baseline. Section 4 studies one margin that bears directly on the OECD bottleneck: redistributing childcare can raise the husband's gain from marriage as income sharing provides less support. Appendix H compares the wage gap with wage levels, fertility preferences, and the division of childcare as determinants of these links.

The policy analysis now asks how particular changes in the economic environment move the three outcomes. I first vary conventional instruments individually, then examine whether coordinating them can produce joint improvements.

3.4 Policy Remedies

I consider three familiar policy objectives in turn: making marriage more attractive, improving women's wage opportunities, and reducing the time cost of children. Each experiment changes one primitive while holding the others fixed. These experiments isolate mechanisms. Applying them to a complete fiscal program would additionally require specifying financing and the other features of the reform.

Pro-marriage policies

Marriage bonuses, joint taxation, and related measures seek to make marriage more attractive, partly because of its connection to family stability and children's outcomes (Kearney, 2023; Autor et al., 2019). I represent their common-benefit component by an increase in λ.

Proposition 1. *An increase in λ raises aggregate fertility n and marriage $\mathcal{M}$ but widens the gender income gap Γ^y.*

The logic is compositional. A higher λ attracts more people into marriage while leaving the equilibrium transfer and fertility within each marital state unchanged. Since married women have more children than single women, average fertility rises as more women marry. Those additional children require maternal time. Female labor supply falls, and the income gap widens. The policy succeeds on marriage and fertility through a channel that works against income equality. Appendix D.6 proves the result.

Gender equality policies

Anti-discrimination and pay-equity measures seek to improve women's labor-market opportunities and reduce misallocation (Hsieh et al., 2019). I isolate their wage effect through an increase in $w^{\female}$ at fixed $w^{\male}$.

Proposition 2. *An increase in $w^{♀}$, holding $w^{♂}$ fixed, reduces aggregate fertility n and marriage $\mathcal{M}$ while narrowing the gender income gap Γ^{y}.*

Higher female wages act through two channels. First, children become more expensive in terms of forgone earnings, reducing fertility among both single and married women. Second, a husband's transfer adds less relative to the wife's own earning capacity, reducing the income-sharing attraction of marriage. Fewer marriages then reinforce the fertility decline.

The transfer share responds to competing forces. Wage convergence and the changing value of fertility exert opposing effects on α^*. Their combined effect unambiguously reduces the transfer relative to women's own earning capacity, $\alpha^*\Gamma^{w}$, and therefore the common marriage premium. Meanwhile, women both earn more per unit of work and spend more time in the market, so the income gap narrows through wages and labor supply together. Appendix D.7 establishes these claims.

Reductions in childcare time costs

A lower time requirement per child appears to offer a direct way to support fertility while preserving women's market work. To examine that possibility, I reduce χ while holding wages and the remaining primitives fixed.

Proposition 3. *A reduction in χ raises aggregate fertility n and marriage $\mathcal{M}$ but widens the gender income gap Γ^{y}.*

Lower childcare costs raise fertility in both marital states. They also increase the equilibrium value of the marital bundle, supporting a larger transfer and more marriage. These direct and composition effects both raise aggregate fertility.

Why, then, does female labor supply fall? The answer lies in the distinction between care per child and total care. With $\rho > 1$, families respond sufficiently strongly to the lower time price that χn rises even though χ falls. Mothers spend less time on each child but more time on childcare altogether. The resulting reduction in market work widens the income gap. Appendix D.8 establishes the response of total childcare time.

Three instruments, three incomplete solutions

Table 5 summarizes the results. Each conventional instrument succeeds on part of the agenda while leaving at least one objective moving in the wrong direction. Pro-marriage and lower-childcare-cost policies raise fertility and marriage at the expense of income equality. Higher female wages narrow the income gap but reduce fertility and marriage.

The pattern in Table 5 raises a natural question: are the individual-instrument trade-offs a fundamental constraint, or an artifact of varying one lever at a time?

Policy	Fertility n	Marriage $\mathcal{M}$	Income gap Γ^y
Pro-marriage ($\uparrow \lambda$)	$+$	$+$	$+$ (widens)
Gender equality ($\uparrow w^{\female}$)	$-$	$-$	$-$ (narrows)
Lower childcare costs ($\downarrow \chi$)	$+$	$+$	$+$ (widens)

Table 5. Effects of Conventional Policies Considered Individually

Notes: Each row changes one instrument while holding the remaining primitives fixed. Comparisons remain within the interior equilibrium region.

Coordinating instruments

The key observation is that the pro-marriage benefit λ and female wages $w^{\female}$ have opposite effects on fertility and marriage. Although a larger marriage benefit widens the income gap on its own, an appropriately sized increase can offset the demographic effects of higher female wages without exhausting their income-gap gains. The following proposition makes this precise.

Proposition 4 (Coordinated instruments)**.** *At an interior baseline equilibrium, a sufficiently small increase in $w^{\female}$ combined with an appropriately chosen increase in λ raises aggregate fertility n and marriage $\mathcal{M}$ while narrowing the gender income gap Γ^y. The child-based family-structure counterpart also improves.*

To see why, raise $w^{\female}$ slightly and adjust λ to hold aggregate fertility fixed. Household fertilities and the transfer do not depend on λ (Lemma 1), while both n^m and n^s fall with the higher wage, so holding n constant requires a shift toward marriage. Female labor supply $1 - \chi n$ is unchanged, so the income gap narrows through the wage alone. A slightly larger increase in λ then raises fertility strictly while, by continuity, preserving the other two improvements. Appendix D.9 gives the proof, including the child-based family-structure measure.

Because the construction applies at every interior baseline equilibrium, the trade-offs in Table 5 reflect varying one instrument at a time rather than a fundamental constraint. Implementation, however, requires matching the two instruments. Too small a marriage-benefit adjustment leaves fertility below its initial level, and too large a one reverses the income-gap gain. This informational demand motivates asking whether a single policy margin can deliver joint improvements.

4 Redistributing Childcare Between Spouses

The baseline assigns all childcare to mothers. Time-use evidence documents a substantial maternal share of care (Aguiar and Hurst, 2007; Kleven et al., 2019), while institutions and policies such as reserved paternity leave can increase fathers' involvement (Farré and González, 2019). Allowing both spouses to provide care introduces a policy margin that changes the distribution of childcare costs within marriage.

Its effects depend on both spouses' budgets and participation incentives. When the husband takes on additional care, the wife gains market time, but his earnings fall and reduce the income available to finance her transfer. The resulting consumption–fertility bundle must also remain attractive to him. I extend the model to incorporate these effects while retaining an equilibrium transfer that clears the marriage market.

4.1 Setup and Household Choices

Let $\chi^{\female}$ and $\chi^{\male}$ denote the *effective* childcare per child provided by wives and husbands:

$$\chi^{\female} \geq 0, \qquad \chi^{\male} \geq 0, \qquad \chi^{\female} + \chi^{\male} = \chi. \tag{12}$$

Single mothers continue to provide χ units of care per child. Marriage thus changes who provides the care while preserving the total effective parental time required for a child.[13]

A parameter $\eta \in (0, 1]$ captures *paternal care efficiency*: one unit of the father's time produces η effective units of childcare, so his raw time per child is $\chi^{\male}/\eta$. When $\eta = 1$ both parents are equally efficient; when $\eta < 1$ the father requires more time to deliver the same effective care.

In this case, the wife's budget becomes

$$c^{\female,m} = \alpha w^{\male}(1 - \tfrac{\chi^{\male}}{\eta} n^m) + w^{\female}(1 - \chi^{\female} n^m),$$

and the husband's budget becomes

$$c^{\male,m} = (1-\alpha) w^{\male}(1 - \tfrac{\chi^{\male}}{\eta} n^m).$$

The transfer remains a share of the husband's earnings, which now depend on his childcare obligations. As before, the wife chooses fertility conditional on the transfer share, and the husband evaluates the resulting bundle.

It is useful to rewrite the wife's budget in terms of effective resources and an effective childcare cost:

$$\tilde{w}^{\female} = w^{\female} + \alpha w^{\male}, \qquad \tilde{\chi}^{\female} = \frac{w^{\female}\chi^{\female} + \alpha w^{\male}\chi^{\male}/\eta}{w^{\female} + \alpha w^{\male}}. \tag{13}$$

Her budget is then $c^{\female,m} = \tilde{w}^{\female}(1 - \tilde{\chi}^{\female} n^m)$. The effective childcare cost is an income-weighted average: a child uses some of her own earning capacity and some of the husband's, the latter adjusted by $1/\eta$ because the father requires $1/\eta$ raw hours per effective unit of care.

[13] Appendix E.6 gives the corresponding analysis when total requirements differ by marital status. Results remain unchanged.

The interior fertility choices are

$$n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^{\female}}{\beta}\right)^{\rho}(\tilde{w}^{\female})^{\rho-1} + \tilde{\chi}^{\female}}, \qquad n^s = \frac{1}{C}. \tag{14}$$

This is the same household trade-off as before, expressed using the resources and fertility price relevant within marriage.

Lemma 4 (Indirect utility with shared childcare)**.** *The wife's marital value is*

$$V^{\female,m} = \lambda(1-\beta)^{1/r}\tilde{w}^{\female}\left[1 + \left(\frac{\beta}{1-\beta}\right)^{\rho}(\tilde{w}^{\female}\tilde{\chi}^{\female})^{1-\rho}\right]^{1/(\rho-1)}. \tag{15}$$

Her single value follows by setting $\lambda = 1$, $\tilde{w}^{\female} = w^{\female}$, *and* $\tilde{\chi}^{\female} = \chi$.

Equations (14) and (15) show why the effective price matters. At fixed resources, a lower price raises fertility and the value of marriage to the wife. With $\rho > 1$, she substitutes sufficiently strongly toward children that consumption falls. Her utility gain comes from the opportunity to choose a more attractive combination of consumption and fertility. Appendix E.1 provides the derivation.

4.2 When Redistribution Improves All Three Outcomes

Consider a local policy that shifts one *effective* unit of childcare per child from the wife to the husband:

$$\frac{d\chi^{\male}}{d\delta} = 1, \qquad \frac{d\chi^{\female}}{d\delta} = -1.$$

Because the father's raw time per child is $\chi^{\male}/\eta$, this shift requires $1/\eta$ additional raw hours from him. When $\eta = 1$ the experiment coincides with a one-for-one time swap; when $\eta < 1$ the father devotes more raw time than the mother saves, reflecting his lower efficiency.

Wages, preferences, and single mothers' childcare costs remain fixed. Total derivatives below include the equilibrium adjustment of α; at $\chi^{\male} = 0$, they are interpreted as right-hand derivatives. Appendix E.2 derives the local equilibrium response formulas used below.

Treating the childcare division as a policy variable captures the possibility that caregiving arrangements adjust more slowly than market opportunities. In Mexico, for example, wages converged substantially over 1990–2025 while the female share of childcare fell only gradually (Section 5). Deschamps et al. (2026) document substantial cross-country variation in child penalties in time use at similar income levels, with that variation tracking cultural proxies more closely than economic fundamentals. The persistence of childcare norms (Goldin, 2025) motivates examining changes in the division of care separately from wages and preferences, whether induced by mandated

paternity leave, workplace regulation, or gradual norm change (Albrecht et al., 2024; Xi and Zhou, 2026).

Three sufficient conditions organize the analysis. Condition 1 asks whether redistribution lowers the wife's effective childcare cost at the initial transfer. Condition 2 asks whether relative earnings converge after fertility and marriage respond. Condition 3 asks whether the adjusted marital contract leaves room for the husband to gain as well. Together, they connect the initial time reallocation to household choices, market participation, and aggregate income.

Condition 1 (the wife faces a lower effective childcare cost)**.** *At the initial transfer, redistribution lowers the wife's effective childcare cost if*

$$w^{♀} > \frac{\alpha w^{♂}}{\eta}.$$

The intuition is straightforward. An effective hour released from the wife's childcare raises her potential earnings by $w^{♀}$. Replacing that effective hour requires $1/\eta$ raw hours of the husband's time, reducing the earnings available to finance her transfer by $\alpha w^{♂}/\eta$. If the first exceeds the second, shifting care toward the husband lowers the price of children from her perspective. Her desired fertility and marital value both rise at the initial transfer.

Condition 1 compares the earning capacities affected by the redistribution, accounting for the father's efficiency. When $\eta = 1$ it reduces to $w^{♀} > \alpha w^{♂}$; at a fixed transfer share, lower η makes the condition harder to satisfy because each effective unit of paternal care consumes more of the father's market time and hence more of the transfer base.

The second condition follows the reform through aggregate childcare. If redistribution raises fertility and marriage, it also creates additional care to be provided. Income convergence depends on how that larger total is divided between mothers and fathers.

Average labor supplies and the income ratio are

$$\begin{aligned} l^{♂} &= 1 - \frac{\chi^{♂}}{\eta}\mathcal{M}n^{m}, \\ l^{♀} &= 1 - \chi n + \chi^{♂}\mathcal{M}n^{m}, \\ \Gamma^{y} &= \Gamma^{w}\frac{l^{♂}}{l^{♀}}. \end{aligned} \tag{16}$$

The father's raw time per child, $\chi^{♂}/\eta$, determines male labor supply. His effective care contribution $\chi^{♂}$ enters female labor supply because each effective unit the father provides is an effective unit the mother does not.

Condition 2 (relative earnings converge after fertility adjusts)**.** *The income-gap condition is*

$$\chi \frac{dn}{d\delta} < \left[\mathcal{M} n^m + \chi^{♂} \frac{d(\mathcal{M} n^m)}{d\delta} \right] \left(1 + \frac{l^{♀}}{\eta \, l^{♂}} \right).$$

The left side is the increase in total childcare generated by higher fertility. The bracket on the right is the increase in fathers' aggregate *effective* childcare, including their care for additional marital births. The final factor accounts for the two-sided effect on relative earnings: each effective unit the father takes on costs him $1/\eta$ raw hours of market time while freeing one hour for the mother. When $\eta < 1$ this asymmetry amplifies the direct male-side adjustment, although the full effect on the condition also depends on equilibrium responses.

Condition 2 requires the male-to-female labor income ratio to decline. This moves incomes toward equality locally when $\Gamma^y > 1$. It can occur even if additional fertility reduces female labor supply, provided male labor supply falls proportionately more. Allowing fathers to provide most childcare also allows the possibility that Γ^y reaches or falls below one. Beyond parity, a further decline in the ratio would no longer represent greater income equality.

The equilibrium response also depends on the husband's valuation of the reform. He loses earnings when he provides more care, but he may value the additional fertility enough to accept that loss. Moreover, the transfer can adjust to redistribute the gains. To express this trade-off, define

$$z \equiv \tilde{\chi}^{♀} n^m \in (0,1), \qquad R \equiv \frac{\beta}{1-\beta} \left(\frac{c^{♂,m}}{n^m} \right)^{1/\rho} - (1-\alpha) w^{♂} \frac{\chi^{♂}}{\eta}.$$

Here z is the share of the wife's effective resources devoted to fertility. The term R measures the husband's marginal value of another child in consumption units, net of the earnings cost associated with his current care obligation. That cost depends on his raw time requirement, $\chi^{♂}/\eta$.

Condition 3 (the husband gains from the adjusted marital contract)**.** *The male-side condition is*

$$\frac{R \left(w^{♀} - \dfrac{\alpha w^{♂}}{\eta} \right) \rho (1-z)}{\tilde{w}^{♀} \tilde{\chi}^{♀}} > \left(1 - \frac{\chi^{♂}}{\eta} \, n^m \right) \left(\frac{w^{♂}}{\eta} - w^{♀} \right).$$

A compensated comparison gives Condition 3 its economic meaning. Consider adjusting the transfer as care is reallocated so that the wife's marriage premium remains at its initial level. Fertility responds to both the change in childcare costs and the compensating transfer. The condition requires the husband to gain along this path.

The left side measures his net valuation of the induced fertility response. The right side captures the earnings cost of moving effective care toward the higher-wage spouse. Each effective

unit requires $1/\eta$ raw hours from the father while releasing one hour of the mother's time, creating the per-unit earnings difference $w^{\male}/\eta - w^{\female}$.

When the inequality holds, the reform leaves a gain for the husband after holding the wife indifferent. The market-clearing transfer can then share that improvement, raising both spouses' marriage premiums. Appendix E.3 derives this compensated comparison and its connection to the equilibrium response.

Proposition 5 (Redistributing childcare)**.** *Consider an interior equilibrium with $w^{\male} > w^{\female} > 0$, $0 < \alpha < 1$, $\eta \in (0, 1]$, and (12). If Conditions 1 and 3 hold, the equilibrium is locally regular and a sufficiently small redistribution toward fathers raises marriage, marital fertility, and aggregate fertility. Both spouses' marriage-to-singlehood utility ratios rise. If Condition 2 also holds, the male-to-female labor income ratio declines, moving incomes toward equality locally whenever $\Gamma^y > 1$. Thus the three conditions together imply*

$$\frac{dn}{d\delta} > 0, \qquad \frac{d\mathcal{M}}{d\delta} > 0, \qquad \frac{d\Gamma^y}{d\delta} < 0.$$

The proof appears in Appendix E.4. The result works through household choices and market participation together. The lower effective price encourages fertility, while the husband's gain from the adjusted bundle supports a higher common marriage premium. More people marry, and married women choose more children. Because single fertility is unchanged and marital fertility is higher, both changes raise aggregate fertility.

Income convergence requires a further step. Additional children bring additional care obligations, and redistribution must be sufficient to improve women's earnings relative to men's after those obligations are taken into account. Conditions 1 and 3 resolve the participation problem; Condition 2 resolves the relative-income problem.

Starting from the baseline, $\chi^{\male} = 0$, the relative-earnings condition simplifies to

$$\chi \frac{dn}{d\delta} < \frac{\mathcal{M} n^m}{\eta} \left(\eta + l^{\female} \right).$$

Appendix E.5 gives the corresponding response formulas using the closed-form baseline equilibrium.

The scope for redistribution also depends on paternal care efficiency. Lower efficiency increases the father's earnings cost per effective unit of care, but its overall effect must account for the resulting changes in fertility, transfers, and relative labor supply. Section 5.4 evaluates this sensitivity by varying η and re-solving the equilibrium while holding wages, preferences, and the effective childcare division fixed.

Proposition 5 thus offers a single-instrument alternative to the coordinated policy in Propo-

sition 4, at the cost of requiring favorable conditions on the wife's effective price, the husband's participation gain, and relative earnings.

The three conditions are sufficient rather than necessary; Appendix E.8 gives the exact first-order criterion for joint improvement. The next subsection asks how far redistribution can proceed while the sufficient conditions continue to hold.

4.3 Limits to Redistribution

As fathers take on more care, fertility and the transfer adjust, changing both spouses' incentives. The local gains in Proposition 5 must therefore be reassessed along the redistribution path. The next proposition bounds the paternal childcare share at which its sufficient conditions can hold jointly.

Proposition 6 (A ceiling on redistribution)**.** *Maintain $\beta \in (0,1)$, $\rho > 1$, $w^{\male} > w^{\female} > 0$, $\eta \in (0,1]$, and a common marriage benefit $\lambda > 0$. Consider interior equilibria with $0 < \alpha < 1$, childcare contributions satisfying (12), and marriage-market clearing as in (4). Let $\bar{\omega}$ be the unique solution in $(\eta/(1+\eta), 1)$ to*

$$\big[(1+\eta)\bar{\omega} - \eta\big]\bar{\omega}^{\rho-1} = \eta^{\rho}(\Gamma^{w})^{1-\rho}. \tag{17}$$

Conditions 1 and 3 can both hold only if

$$\frac{\chi^{\male}}{\chi} < \bar{\omega} < 1.$$

At any interior equilibrium satisfying the maintained assumptions with $\chi^{\male}/\chi \geq \bar{\omega}$, Condition 1 implies $R < 0$, so Condition 3 fails. All three sufficient conditions in Proposition 5 therefore require a paternal childcare share below $\bar{\omega}$.

The proof appears in Appendix E.7. The intuition combines household choice with marriage-market clearing. Under Condition 1, marriage gives the wife greater full resources and a weakly lower price of fertility than singlehood. After removing the common marriage benefit, she therefore enjoys a utility gain from marriage. Market clearing requires the husband to enjoy the same proportional gain. Once his childcare share is sufficiently high, that requirement is incompatible with a nonnegative net marginal value of additional fertility, $R \geq 0$. The male-side condition then fails even though shifting care toward him would still lower the wife's effective childcare cost at the initial transfer.

With equally efficient parents, the ceiling satisfies

$$(2\bar{\omega} - 1)\bar{\omega}^{\rho-1} = (\Gamma^{w})^{1-\rho}, \qquad \frac{1}{2} < \bar{\omega} < 1.$$

The ceiling therefore lies above equal sharing when $\eta = 1$.

Because the ceiling restricts only Conditions 1 and 3, the conditions may fail jointly well before $\bar{\omega}$.[14] Locating that first failure requires re-solving fertility and the transfer as care is redistributed. Section 5.4 does so for each year of Mexico's transition.

4.4 Relationship to the Empirical Evidence

Reserved paternity leave provides a natural setting for studying changes in fathers' caregiving. Evidence from Spain (Farré and González, 2019), Norway (Cools et al., 2015), Denmark (Kleven et al., 2026), South Korea (Lee and Lee, 2025), and Quebec (Patnaik, 2019) examines how reforms affect the household division of labor. This evidence addresses the first practical requirement of the redistribution experiment: policy must increase fathers' contribution to care. The literature generally finds that it does.

The next question concerns fertility and family formation. Fertility responses differ across settings. Subsequent fertility falls following the Spanish reform studied by Farré and González (2019), while Lee and Lee (2025) find an increase following paternal leave policies in South Korea. The model organizes this heterogeneity through the effects of care reallocation on mothers' effective costs, fathers' consumption, and the equilibrium transfer. Similar changes in caregiving can produce different responses when preferences, wages, and initial allocations differ. This motivates a setting-specific quantitative assessment such as the Mexico calibration in Section 5.

A final question is whether earnings converge after families adjust. Higher maternal employment is one route to convergence, but the model also allows the income ratio to fall when men's earnings decline proportionately more than women's. Evaluating the full mechanism therefore requires information on both parents' wages and labor supply, fertility by marital status, transfers, and marriage participation.

4.5 Public Childcare and the Marketization of Childcare

Public childcare and purchased care provide another way to release parental time. Outsourcing can reduce total parental care per child, introduce a monetary payment, and change the two parents' remaining obligations differently. Its effects therefore combine several mechanisms.

The baseline reduction in χ isolates a lower maternal time requirement while holding additional spending and financing outside the experiment. Proposition 3 shows that, under $\rho > 1$, the fertility response is strong enough for total maternal childcare to rise. This result highlights the distinction between time saved per child and total time saved after fertility adjusts. Applying it to

[14] Starting from an allocation where all three hold, their first failure must occur before an admissible interior path reaches $\bar{\omega}$, unless the path first reaches an equilibrium boundary. The argument does not require the conditions to deteriorate monotonically.

public childcare would require accounting for the reform's other effects on household resources and choices.

The redistribution experiment focuses on who provides the remaining care. Moving an effective hour from the wife to the husband lowers his earnings by $w^{\male}/\eta$ as it releases $w^{\female}$ of hers. Purchased childcare replaces parental time with a service whose cost must be financed. The associated budgets and participation incentives differ, so extending Proposition 5 to outsourced care requires incorporating those payments and their incidence.

These distinctions reinforce the need to follow a reform through both household budgets and the marriage market. Its effects depend on the wife's effective price of children, the husband's valuation of the adjusted marital bundle, and the allocation of care after fertility responds. Extending the model to purchased childcare would require specifying service prices, financing, and substitution with parental time. The quantitative analysis that follows focuses on the parental redistribution mechanism developed here.

5 Quantitative Assessment

I calibrate the shared-childcare model to Mexico from 1990 to 2025 to assess the mechanisms developed above. Mexico combines a long demographic transition with substantial wage changes and survey evidence on the division of childcare. The analysis proceeds from data and calibration to equilibrium transfers and welfare, then evaluates the scope for further redistribution. A final decomposition separates the contributions of productivity, relative wages, and childcare allocation to the observed transition.

5.1 Data

Core variables

The calibration uses three core series. *Fertility* is the total fertility rate from the United Nations. *Dual parenthood* is children's co-residence with both parents, from Brenøe and Wasserman (2025).[15] The *female labor income share* is from the World Inequality Database.

[15]The model's marriage margin represents a co-resident two-parent arrangement, including both legal marriage and consensual unions. The calibration target measures children's co-residence with both biological parents and therefore includes *uniones libres*, which are widespread in Mexico and grew over the sample period. This broader partnership concept matches the model's income-sharing and childcare-sharing mechanisms. The dual-parenthood series consequently avoids treating substitution between legal marriage and cohabitation as a loss of two-parent co-residence.

Additional data

Two additional data series, not used in the cross-country analysis, are required for the extended-model calibration.

GDP per capita in constant 2010 U.S. dollars is from the Federal Reserve Economic Data (FRED). Over 1990–2025, Mexican GDP per capita rises from roughly \$8,000 to approximately \$10,300, with substantial volatility around the 1994–95 Peso Crisis, the 2008–09 recession, and the 2020 pandemic downturn.

The gender division of childcare is constructed from micro-level data from Mexico's National Survey on Time Use (Encuesta Nacional sobre Uso del Tiempo, ENUT). Three survey waves provide anchor points for the female share of total parental childcare time:

- 1996 (ENTAUT): female share $= 0.85$;
- 2009 (ENUT, Cuadro 3.1.1): female share $= 0.80$;
- 2024 (ENUT): female share $= 0.73$.

A piecewise-linear interpolation through the three anchors generates an annual series. The resulting path declines at approximately -0.38 percentage points per year before 2009 and -0.47 percentage points per year thereafter, reflecting the gradual shift of childcare responsibilities toward fathers documented in the time-use literature.

Figure 3 plots the five series used in the calibration.

5.2 Calibration

The series describe a transition toward lower fertility, less widespread dual parenthood, and a higher female income share. Fertility falls from approximately 3.4 to 1.8 children per woman. Children's co-residence with both parents declines from 81 to approximately 78 percent, with a dip below 75 percent around 2017. The female income share rises from 22 to 34 percent, while real GDP per capita increases from roughly \$8,000 to over \$10,000 in 2010 dollars. The calibration asks how much of this joint movement the model can reproduce using changes in productivity, relative wages, and childcare allocation.

Drivers

The extended model requires three time-varying inputs in each year t:

1. **Gender-neutral productivity**, A_t, which equals the male wage $w_t^{\male}$.

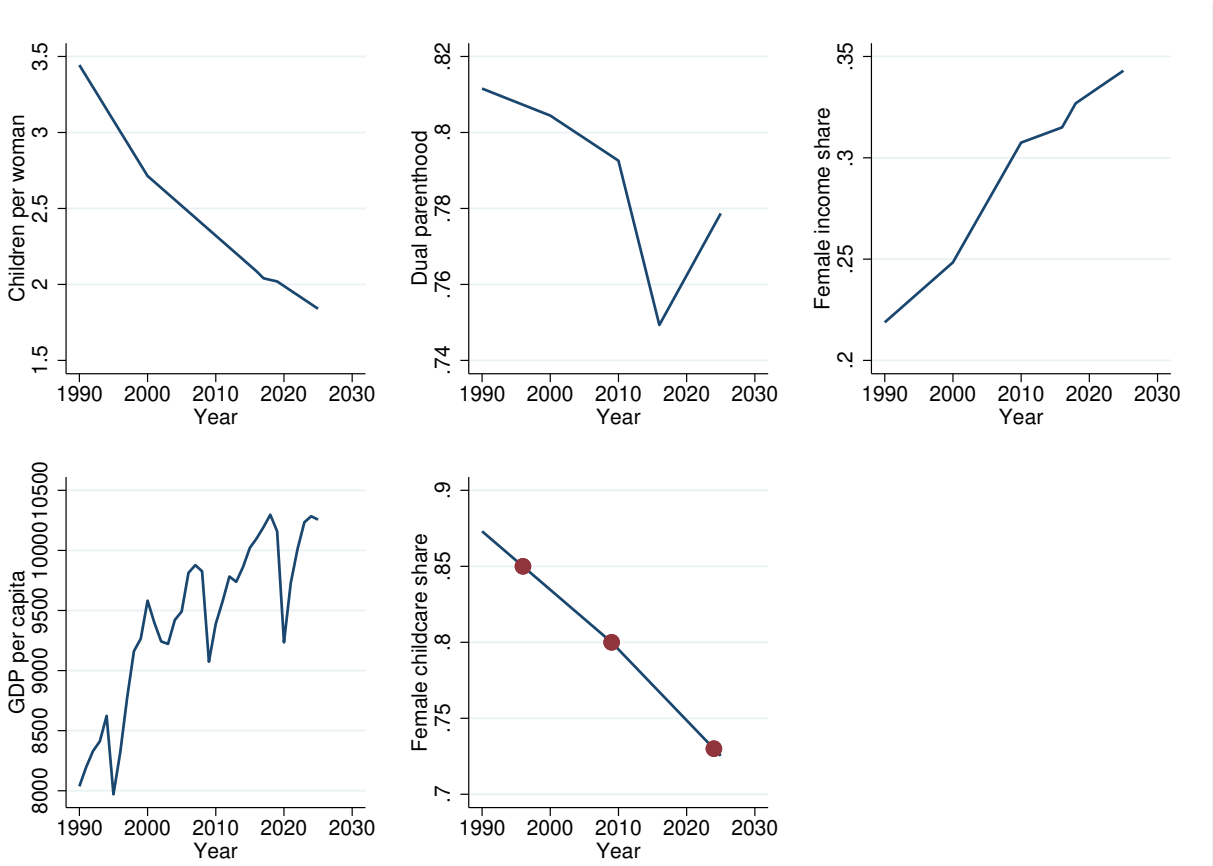


Figure 3. Mexico's Transition Path (1990–2025)

Notes: Children per woman from the United Nations. Dual parenthood measures children's co-residence with both parents (Brenøe and Wasserman, 2025). Female labor income shares from the World Inequality Database. GDP per capita in constant 2010 U.S. dollars from FRED. Female childcare share from a piecewise-linear interpolation through ENUT survey anchors (dots) in 1996, 2009, and 2024.

2. **The relative wage gap**, $B_t \equiv w_t^{\male}/w_t^{\female}$, which captures all gender-biased factors including human capital differences, discrimination, and occupational sorting.

3. **The gender division of childcare**. The ENUT-based series provides the population-wide female share of total parental childcare time, s_t. Because single mothers provide all childcare (χ per child), the within-marriage female share φ_t satisfies

$$s_t = \frac{\mathcal{M}\varphi_t \chi n^m + (1-\mathcal{M})\chi n^s}{\chi n},$$

so $\varphi_t < s_t$ whenever some women are single mothers. Given s_t and the model's equilibrium $(\mathcal{M}, n^m, n^s, n)$, the within-marriage shares are $\chi_t^{\female} = \varphi_t \chi$ and $\chi_t^{\male} = (1-\varphi_t)\chi$. The recovery is iterated jointly with the equilibrium solver until convergence.

Throughout the calibration, paternal care efficiency is set to $\eta = 1$, so that one hour of the father's time produces one effective hour of childcare. Section 5.4 examines robustness to $\eta < 1$.

Recovering wages

I set $\chi = 0.075$ following De La Croix and Doepke (2003).[16] The wage-recovery procedure uses observed fertility and the childcare-share series to construct female and male labor supplies:

$$l_t^{\female} = 1 - s_t \chi n_t, \qquad l_t^{\male} = 1 - (1 - s_t)\chi n_t,$$

where n_t is observed fertility.[17] Combining these constructed labor supplies with GDP per capita and the female income share gives the wage-recovery identities

$$w_t^{\male} = \frac{2\tilde{A}_t(1-\sigma_t)}{l_t^{\male}}, \qquad w_t^{\female} = \frac{2\tilde{A}_t\sigma_t}{l_t^{\female}},$$

where $\tilde{A}_t$ is GDP per capita normalized to its 1990 value and σ_t is the female income share. These identities yield the time-varying inputs $A_t = w_t^{\male}$ and $B_t = w_t^{\male}/w_t^{\female}$ used in the equilibrium calculations.

Preference and marriage parameters

Four parameters $\{\beta, \rho, \lambda, \theta\}$ are chosen to match the mean and change in fertility and dual parenthood over 1990–2025. The model's marriage rate and relative fertility are mapped into the child-based family-structure measure using Appendix D.4. The parameters have different primary roles: β governs the level of fertility, ρ its response to productivity, λ the common benefit of marriage, and θ participation responsiveness. Because household choices and marriage are jointly determined, each parameter can affect several targets. Table 6 reports the estimates.[18]

The estimate $\rho = 3.0$ places the economy in the substitution-dominant region ($\rho > 1$). The common marriage benefit, $\lambda = 0.945$, is below unity, but combines with the income-sharing component to support the observed family-structure outcomes. Participation responsiveness is high ($\theta = 18$), so modest changes in the marriage-to-singlehood utility ratio can produce substantial changes in marriage.

[16] Appendix G repeats the calibration for $\chi \in \{0.05, 0.10\}$ and examines the sensitivity of the main results.

[17] These aggregate labor supplies use the population-wide share s_t rather than the within-marriage share φ_t because $s_t\chi n_t$ equals total female childcare time per woman across both marital states, and analogously for men.

[18] Appendix C examines local identification through the elasticity Jacobian of the four targeted moments with respect to $(\beta, \rho, \lambda, \theta)$ and its inverse. The forward matrix shows how parameters affect the targets, while the inverse reports how parameter estimates respond to changes in those targets.

	Interpretation	Value	Target	Source
χ	Time cost of children	0.075	De La Croix and Doepke (2003)	
η	Paternal care efficiency	1	Baseline (robustness in Section 5.4)	
A_t	Gender-neutral productivity		GDP per capita	FRED
B_t	Relative wage gap ($w^{\male}/w^{\female}$)		Female labor income share	World Inequality Database
s_t	Female childcare share		ENUT anchors	ENUT 1996, 2009, 2024
β	Fertility preference weight	0.068	Fertility level	United Nations
ρ	Consumption–fertility substitutability	3.0	Fertility trend	United Nations
λ	Common marriage benefit	0.945	Dual-parenthood level	Brenøe and Wasserman (2025)
θ	Participation responsiveness	18.0	Dual-parenthood trend	Brenøe and Wasserman (2025)

Table 6. Calibration Results

Notes: Time-varying inputs are constructed from data. Preference and marriage parameters are chosen to match the model's fertility and children's co-residence with both parents to the data over 1990–2025. The gender division of childcare is pinned by three ENUT survey waves.

Model fit

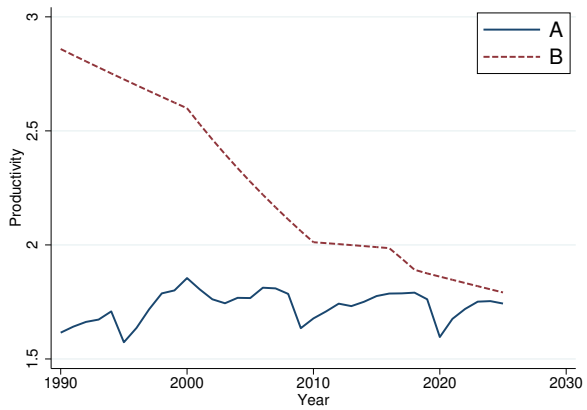
Figure 4 displays the recovered productivity inputs and the model fit. Gender-neutral productivity, A_t, fluctuates modestly around 1.7, while the relative wage gap, B_t, declines from approximately 2.9 to 1.8. The model tracks the fertility decline with a correlation of 0.93 and a root-mean-square error of 0.18 children per woman. For dual parenthood, the correlation is 0.29 and the RMSE is 0.03. The model captures its level and broad downward movement, though the short-run variation in dual parenthood is less well reproduced.

Non-targeted validation: female labor force participation

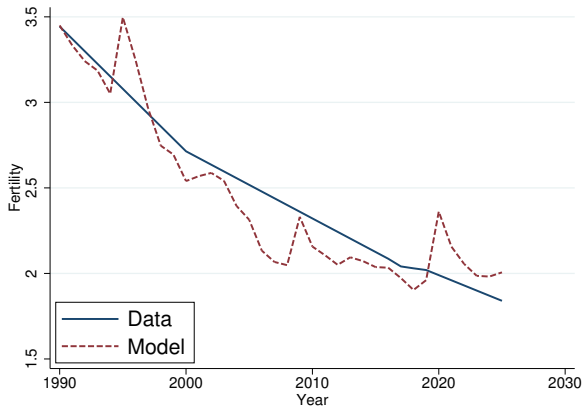
The calibration targets fertility and dual parenthood, while observed female labor force participation (FLFP) is not used as an estimation target. I compare the model-implied female labor supply, $l_t^{\female} = 1 - \chi s_t n_t$, with Mexico's FLFP series from the Federal Reserve Economic Data (FRED). This comparison examines whether the fitted demographic transition is accompanied by a plausible pattern of women's market work.

The two series measure different margins. Model labor supply is an intensive-margin time allocation, ranging from approximately 0.78 to 0.90 over the sample, whereas FLFP measures extensive-margin participation in percentage points. I therefore normalize each series by its total 1990–2025 change, setting both to zero in 1990 and one in 2025. This preserves the share of the total change accumulated at each date and permits a comparison of the timing and shape of the transition.

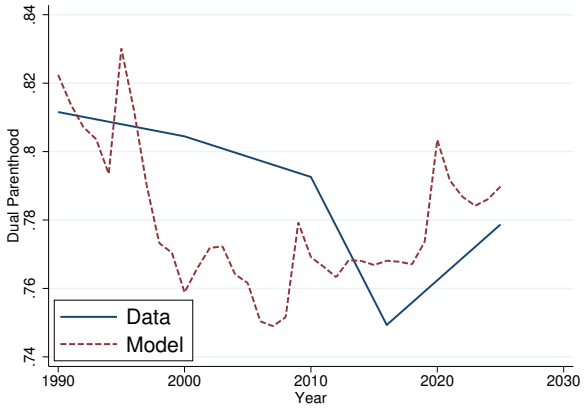
Figure 5 shows that the normalized series have a correlation of 0.976. The model reproduces the rapid increase through approximately 2008 and the subsequent deceleration. In the model, the early increase accompanies the sharp decline in the relative wage gap B_t, which raises the



(a) Productivity Inputs (A_t, B_t)



(b) Fertility



(c) Dual Parenthood

Figure 4. Model Fit: Mexico (1990–2025)

Notes: Solid lines denote data and dashed lines denote model values. Panel (a) shows the two productivity inputs: A_t (gender-neutral, solid) and B_t (relative wage gap, dashed). Fertility is children per woman. Dual parenthood measures children's co-residence with both parents; the model maps adult marriage and relative fertility into this child-based measure. Parameters appear in Table 6.

opportunity cost of children and releases time for market work. The later slowdown coincides with more gradual wage convergence. The close correspondence with the non-targeted FLFP series provides a complementary check on the timing of the labor-supply mechanism, while the difference between hours and participation limits a direct comparison of levels.

5.3 Equilibrium Outcomes and Welfare

The fitted equilibrium also recovers within-marriage transfers and systematic welfare by agent type. These objects clarify how wage convergence changes the marital contract and the distribution of gains, even when they are difficult to observe directly.

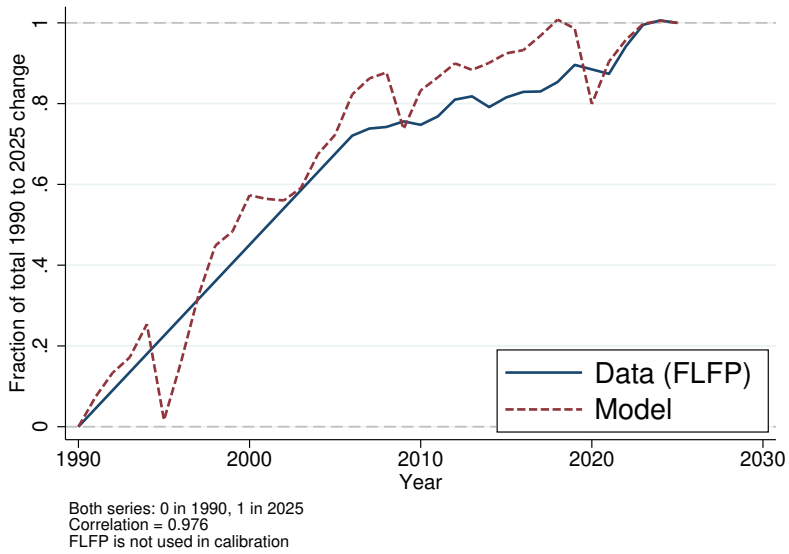


Figure 5. Non-Targeted Validation: Female Labor Supply vs. FLFP

Notes: Both series are normalized as fractions of their respective total 1990–2025 changes (0 in 1990, 1 in 2025). The solid line is Mexico's female labor force participation rate from FRED. The dashed line is the model's predicted female labor supply, $l_t^{\varphi} = 1 - \chi s_t n_t$. FLFP is not used in calibration. Correlation $= 0.976$.

Within-marriage transfers

Figure 6a shows that the equilibrium transfer share α_t falls from 3.1 percent of the husband's earnings in 1990 to 1.2 percent in 2025 (Table **??**).[19] The decline is not monotone: most of it occurs between 1995 and 2000, and the share rises slightly around 2005–2010 and again in 2020. Measured relative to the wife's own wage, the transfer falls more steeply, from 8.8 to 2.2 percent, because the wage gap B_t narrows at the same time. In the baseline model, this ratio is the income-sharing component of women's marriage gains (Lemma 2). Its decline records the shrinking income-sharing support for marriage as wages converge. The cash transfer understates husbands' contribution, however: husbands bear most of the cost of their own childcare time, $(1-\alpha_t)\, w_{m,t}\, \chi_{m,t}\, n_t^M$, which lowers the wife's effective price of children. Valued at the husband's wage, this in-kind contribution rises from about 4 to 8 percent of his potential earnings over the same period, so part of the fall in α_t is a shift from cash to care.

[19] α_t should not be read as the wife's share of household resources, the object estimated in the collective-household literature. It is the payment that clears the marriage market. Because taste shocks enter multiplicatively, market clearing equates the spouses' *proportional* gains from marriage, not their levels. Since the husband earns B_t times the wife's wage, a transfer of α_t of his earnings raises her resources by $\alpha_t B_t$. In addition, children are the only source of marital surplus in the model, so the transfer is bounded by the husband's willingness to pay for them. The model's implied split of household consumption is summarized by the consumption equivalents in Table **??**.

Consumption equivalents

I measure welfare using consumption equivalents benchmarked to the married male. For each agent type j, the consumption equivalent CE_j is the proportional increase in consumption that agent j would need, holding fertility and marital status fixed, to achieve the married male's utility level. Formally, $\mathrm{CE}_j = \mu_j - 1$, where μ_j solves $u(\mu_j c_j, n_j) = V^{\male,m}/\lambda$ when j is married (using the systematic component) and the analogous expression for single agents. A positive consumption equivalent indicates that the agent is worse off than the married male and would require additional consumption to match his welfare; a negative value would indicate the reverse.[20]

Figure 6b reports positive consumption equivalents for all three non-benchmark agent types throughout the sample. Under this systematic-utility comparison, each would require additional consumption to attain married-male welfare. All three measures decline over time, indicating convergence toward that benchmark.

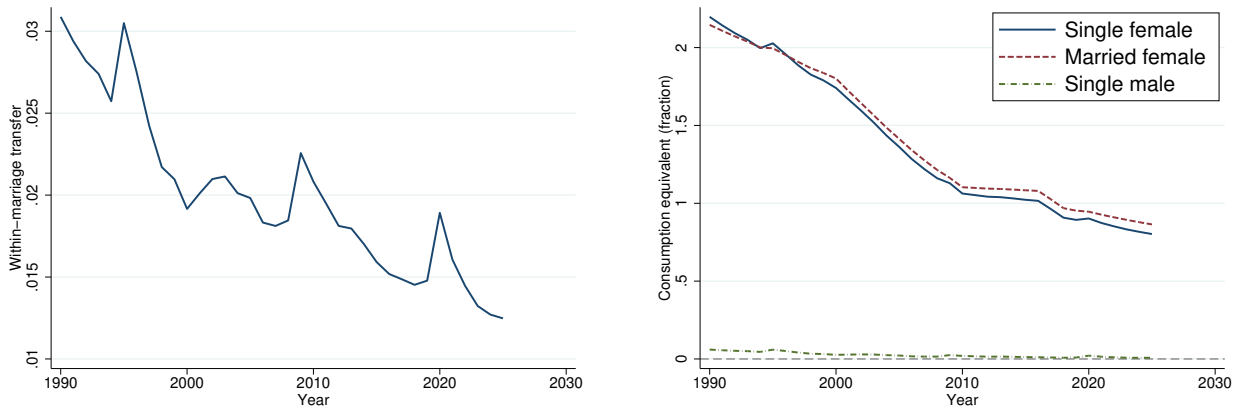
Single women have the largest initial gap. Their consumption equivalent is 2.20 in 1990, meaning that matching married-male welfare would require 220 percent more consumption at their existing fertility. By 2025, the measure falls to 0.80. The married-female consumption equivalent declines from 2.15 to 0.86. Wage convergence drives most of these changes (Section 5.5). Because the transfer relative to the wife's wage falls over the same period, the convergence reflects women's own earning power rather than greater income sharing. Single men have the smallest gap, falling from 0.06 in 1990 to 0.01 in 2025, with most of the decline occurring by 2000. Because a single man's value is $u(w^{\male}, 0)$, his consumption equivalent equals the common marriage premium minus one, $V^{\male,m}/V^{\male,s} - 1$. Its decline is therefore the shrinking marriage premium that, through (3), lowers marriage participation.

5.4 Childcare Redistribution Conditions

The calibrated equilibrium allows three related assessments of the redistribution mechanism. I first evaluate whether a small reform improves all three outcomes at each year's allocation. I then increase the paternal childcare share to locate the first failure of the sufficient conditions. Finally, I return to the observed effective-care division and examine robustness to lower paternal efficiency. Because fathers provide positive childcare throughout ($\chi_t^{\male} > 0$), all three exercises use the general equilibrium derivatives in Appendix E.2.

At equal parental efficiency ($\eta = 1$), all three conditions hold in every year from 2001 to 2025. Condition 1 ($w^{\female} > \alpha w^{\male}$) holds comfortably throughout the full sample: the transfer share never exceeds 3.1 percent of male earnings, so the wife's wage exceeds the transfer-weighted male wage.

[20] For the married female, the consumption equivalent simplifies to $\mathrm{CE}^{\female,m} = c^{\male,m}/c^{\female,m} - 1$, since married spouses share the same fertility. For single agents, the comparison accounts for different fertility levels using the CES aggregator.



(a) Within-Marriage Transfer (α_t)

(b) Consumption Equivalents

Figure 6. Equilibrium Outcomes and Welfare

Notes: Panel (a) shows the equilibrium transfer share α_t. Panel (b) reports consumption equivalents benchmarked to the married male. $\mathrm{CE} > 0$ indicates the agent is worse off than the married male and needs additional consumption to match his welfare. The dashed horizontal line at zero marks the married-male benchmark.

Condition 3 holds throughout because the husband's valuation of the induced fertility response exceeds the earnings cost along the wife-compensated path. Condition 2 (income convergence), however, fails in 1990–2000, when the wage gap is largest and the fertility response to redistribution generates additional childcare obligations that prevent relative-income convergence. The narrowing wage gap after 2000 helps satisfy this condition by moderating the earnings cost of fathers' care.

Proposition 5 therefore predicts gains from a sufficiently small further redistribution at each year's calibrated equilibrium from 2001 onward. The ENUT-based series places the population-wide paternal share at approximately 13 percent in 1990 and 27 percent in 2025; the recovered within-marriage paternal share is larger, rising from approximately 15 to 35 percent over the same period, because single mothers provide all childcare. The next exercise asks how far the within-marriage share can rise before the sufficient conditions first fail.

How far can redistribution proceed?

The analytical ceiling in Proposition 6 provides a general upper bound on the paternal share at which the sufficient conditions can hold. To locate their first failure in Mexico, I hold each year's recovered wages and the calibrated parameters $(\beta, \rho, \chi, \lambda, \theta)$ fixed and maintain $\eta = 1$. Starting from the observed within-marriage paternal share $\omega_t^{\mathrm{obs}} = 1 - \varphi_t$, where φ_t is the within-marriage female share recovered in the calibration, I increase $\omega = \chi^{♂}/\chi$, setting $\chi^{♂} = \omega\chi$ and $\chi^{♀} = (1-\omega)\chi$. Single mothers' childcare requirement remains unchanged.

At each allocation, I re-solve the marriage market and evaluate Conditions 1, 2, and 3 using the equilibrium derivatives in Appendix E.2. The numerical boundary is the first detected paternal

share at which at least one of these strict inequalities ceases to hold along the equilibrium branch starting from the observed allocation.[21] For comparison, I also compute the year-specific analytical ceiling $\bar{\omega}_t$ from (17), evaluated at $\eta = 1$ and $\Gamma_t^w = B_t$.

Figure 7 compares the resulting numerical boundary with the observed allocation and the analytical ceiling. In the early years of the sample, the sufficient conditions already fail at the observed paternal share, so no boundary can be located. From 2001 onward, the numerical boundary rises from approximately 24 percent to 35 percent in 2025. It remains above the observed paternal share throughout these years, though the margin narrows toward the end of the sample. The distance between the numerical boundary and the observed share is roughly 2.1 percentage points in 2001, widens to about 5 percentage points around 2009, and narrows to approximately 0.2 percentage points by 2025. Thus, the observed increase in fathers' contribution is accompanied by continued room for further redistribution under the sufficient conditions, although that room diminishes as the observed share approaches the boundary.

The analytical ceiling rises from approximately 67 percent to 77 percent over the same period, well above the numerical boundary. The general bound therefore leaves substantial room in which a particular economy's conditions may already have failed. In Mexico, the first detected failure occurs below 50 percent in every year where a boundary is located. Along the evaluated equilibrium paths, all three sufficient conditions consequently cease to hold before care is equally shared.

Two qualifications apply to this boundary and to the efficiency threshold below. First, the conditions are sufficient: past the boundary, joint improvement is no longer *guaranteed* by Proposition 5, but further redistribution need not worsen any outcome. Second, the boundary is local. It refers to the equilibrium branch traced from the observed allocation, and it does not rule out gains at other allocations or from a finite reform. With these qualifications, the rising boundary shows that the calibrated economy can sustain a larger paternal contribution as wage convergence lowers the relative earnings cost of fathers' care.

Paternal efficiency threshold

The share-boundary exercise holds parental efficiency fixed and varies the allocation of care. I now hold each year's effective-care allocation fixed and vary paternal efficiency. This complementary experiment asks how much additional raw time fathers can require before the sufficient conditions cease to hold.

To answer it, I conduct the following counterfactual: holding the calibrated preference and

[21] The search increases the paternal share in increments of 0.001, equivalent to 0.1 percentage point, using the preceding equilibrium transfer as the starting value for the next solution. Once a condition failure is detected, the interval between the last passing and first failing allocations is refined to a width below 10^{-6} in share units. The figure reports the failing endpoint. A condition-failure boundary is located in every year from 2001 to 2025.

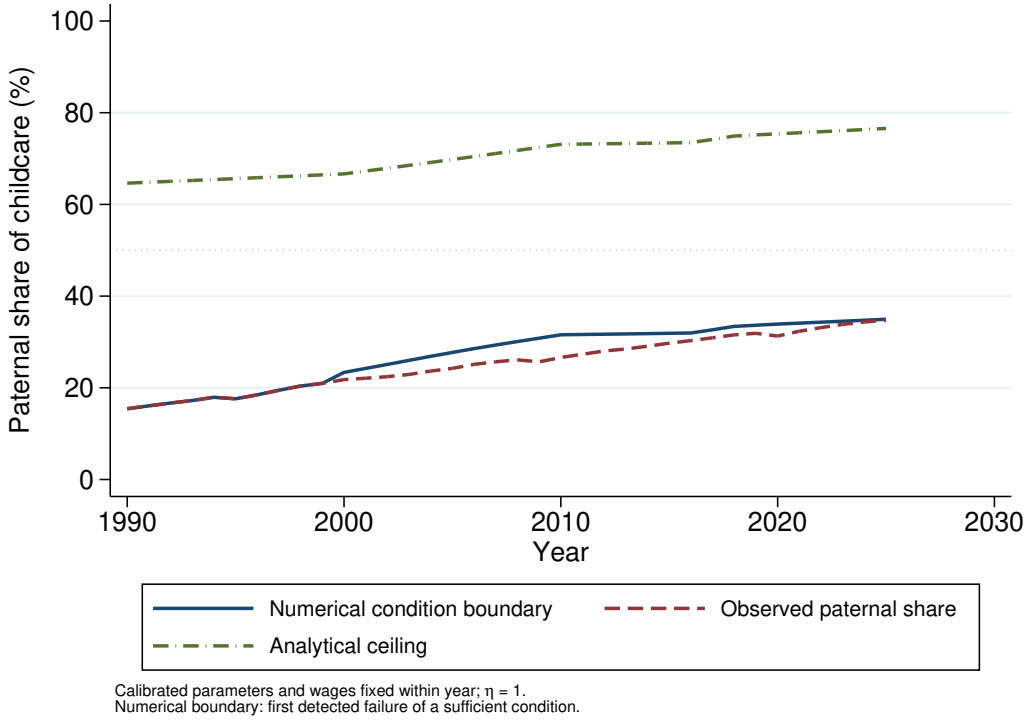


Figure 7. Limits to Childcare Redistribution in Mexico (1990–2025)

Notes: The solid navy line reports the first detected paternal childcare share at which at least one of Conditions 1, 2, and 3 fails along the equilibrium branch starting from each year's observed allocation. Recovered wages and calibrated parameters are held fixed within year, including $\eta = 1$, while fertility and transfers adjust as childcare is redistributed. The dashed maroon line is the observed paternal share constructed from the ENUT-based series. The dash-dot green line is the analytical ceiling from Proposition 6. The horizontal dotted line marks equal sharing. Shares refer to childcare within marriage; with $\eta = 1$, effective-care and raw-time shares coincide.

marriage parameters $(\beta, \rho, \lambda, \theta)$ fixed at their $\eta = 1$ estimates (Table 6), I vary η on a fine grid in $(0, 1]$, set the father's raw time to $\chi_t^{\male}/\eta$, re-solve the equilibrium at each year's recovered wages and effective childcare division, and evaluate the three conditions of Proposition 5.[22]

For each year, define the *paternal efficiency threshold* as the lower boundary of the efficiencies at which all three conditions hold along the equilibrium branch evaluated in the exercise:

$$\bar{\eta}_t \equiv \inf \left\{ \eta \in (0, 1] : \begin{array}{l} \text{the equilibrium is interior with } 0 < \alpha < 1, \\ \text{and Conditions } 1, 2, 3 \text{ all hold} \end{array} \right\}. \tag{18}$$

All equilibrium quantities and policy derivatives are recomputed at each η.

In 1990–2000, no efficiency level in $(0, 1]$ satisfies all three conditions, because Condition 2 already fails at $\eta = 1$; Figure 8 plots $\bar{\eta}_t = 1$ in these years by convention. From 2001 onward, $\bar{\eta}_t$

[22] Because the preference and marriage parameters are held at their $\eta = 1$ values, the exercise isolates the effect of lower paternal efficiency on the redistribution conditions while keeping the rest of the economic environment fixed. A full re-calibration for each η would change the estimated parameters and is left for future work.

declines to approximately 0.81 by 2010, corresponding to a paternal efficiency shortfall of roughly 19 percent. It subsequently rises as the wage gap continues to narrow and the observed paternal share approaches the numerical condition boundary. Wage convergence reduces the relative cost of the father's forgone earnings, $w^{♂}/\eta$, and helps relax Condition 3. Condition 3, the husband's compensated participation gain, is binding at the reported efficiency boundary throughout the years where a finite threshold exists. Condition 1 remains far from binding because the equilibrium transfer share is small.

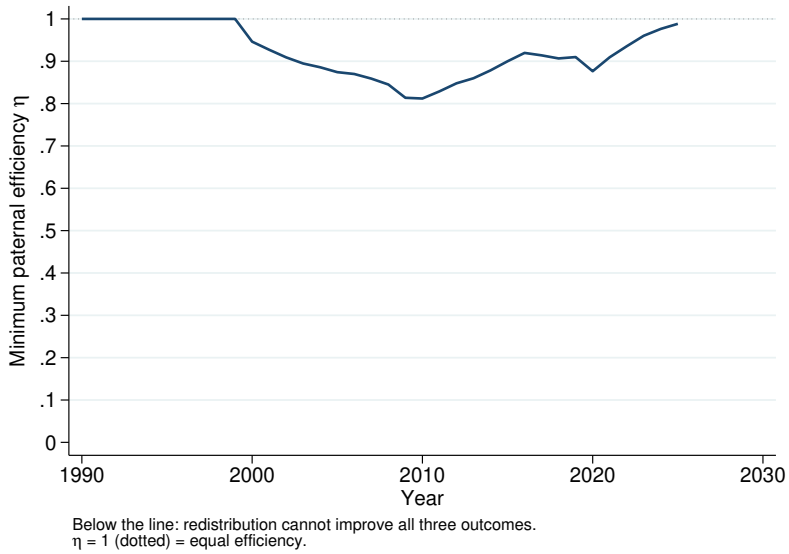


Figure 8. Paternal Efficiency Threshold $\bar{\eta}_t$ (Mexico, 1990–2025)

Notes: The solid line approximates the paternal efficiency boundary $\bar{\eta}_t$ defined in (18), using the smallest grid value at which all three conditions of Proposition 5 hold. The exercise holds calibrated parameters at their $\eta = 1$ estimates (Table 6), fixes each year's recovered wages and effective childcare division, and re-solves the equilibrium as η varies. Years with no passing grid value are plotted at one.

Taken together, the two counterfactuals show that the sufficient-condition region expands as the wage gap narrows during the middle of the sample, with the paternal share at first failure rising and the lower paternal-efficiency boundary falling. Both patterns accompany wage convergence in the calibrated transition. Appendix G shows that the declining efficiency boundary during the 2000s persists under alternative childcare time costs χ.

5.5 Shapley-Value Decomposition

The preceding counterfactuals evaluate redistribution within each year's economic environment. I now separate the forces that move that environment over time. A Shapley-value decomposition attributes changes in fertility, dual parenthood, the gender income gap, and consumption-equivalent welfare to gender-neutral productivity (A), the relative wage gap (B), and the division of childcare

(χ). Here χ labels the childcare-division force; the total time requirement per child remains fixed.

Methodology

For each force, "on" means the variable takes its observed time-varying value; "off" means it is held fixed at its 1990 value. All $2^3 = 8$ combinations are solved. The Shapley formula assigns each force its average marginal contribution across all possible orderings of the three forces. The decomposition is exact: the three Shapley contributions sum to the total change from the all-off baseline in every year. Unlike a sequential decomposition, the Shapley value is invariant to the ordering of forces.[23]

Results

Figure 9 presents the decomposition for several key outcome variables. Table 7 reports the Shapley shares at the end of the sample.

Outcome	A (%)	B (%)	χ (%)
Fertility	30.6	172.0	−102.6
Dual parenthood	94.5	367.6	−362.1
Gender income gap	3.9	94.4	1.7
CE: Single female	3.0	95.5	1.5
CE: Married female	2.4	113.6	−16.1
CE: Single male	22.4	69.5	8.1

Table 7. Shapley Shares at 2025 (% of Total Change from 1990 Baseline)

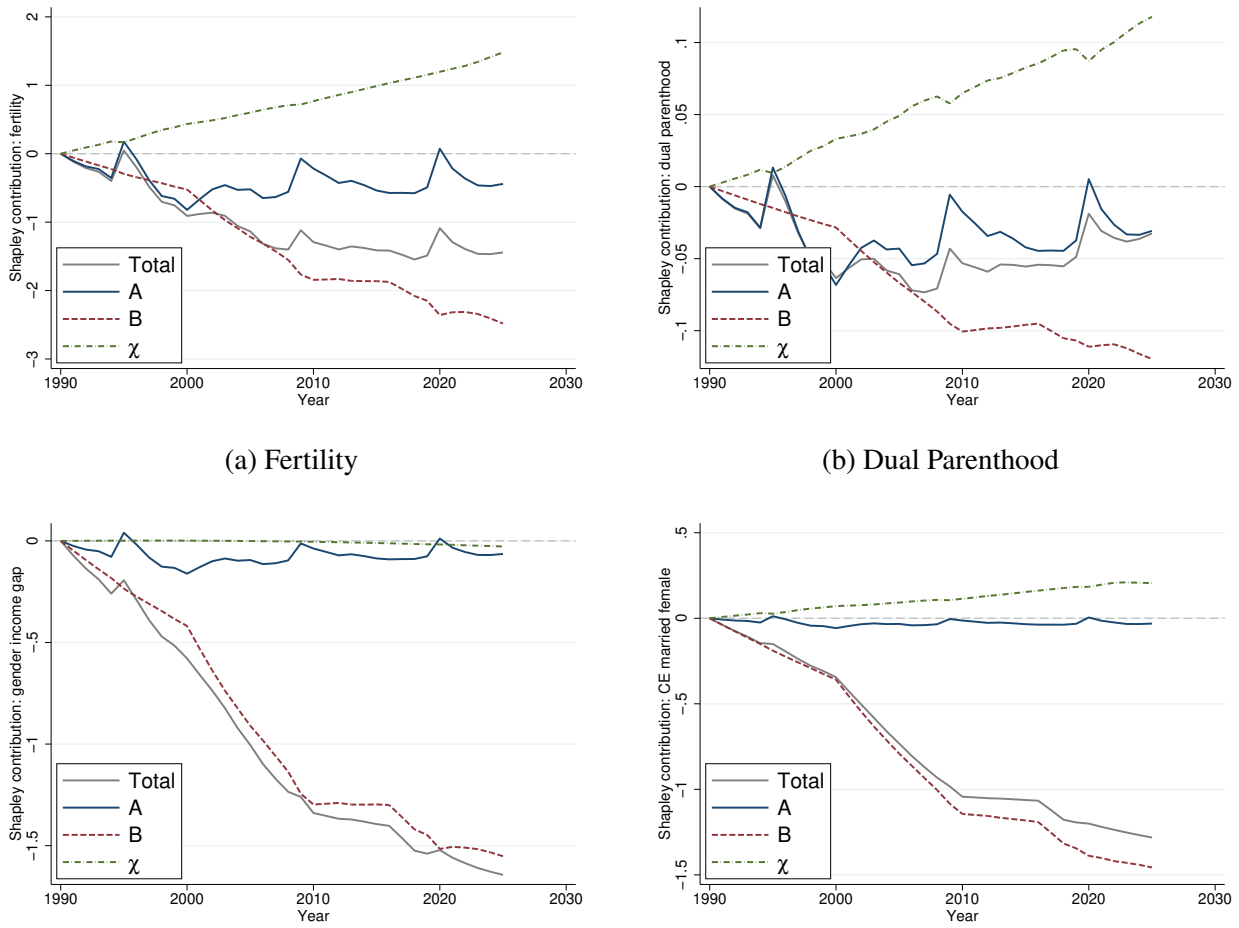
Notes: Each row reports the Shapley share of the three forces as a percentage of the total change from the baseline (all inputs at 1990 values). Shares sum to 100 percent in each row up to rounding. A negative share indicates that the force moves the outcome in the opposite direction from the total change.

Three findings summarize the decomposition.

Wage convergence accounts for most of the transition. The relative wage gap B has the largest contribution for every reported outcome. It explains 172 percent of the fertility decline, 94 percent of income convergence, and 95–114 percent of the decline in female consumption equivalents. Its fertility contribution exceeds the total decline because changes in childcare allocation partly offset it. Higher female wages raise the opportunity cost of children, reduce the relative value of income sharing within marriage, and narrow the income gap directly.

More equal childcare buffers the demographic declines. The childcare-division contribution is -103 percent for fertility and -362 percent for dual parenthood. Because both outcomes

[23] For three forces with scenario values $V_0, \ldots, V_7$, the Shapley value of force A is $\phi_A = \frac{1}{3}(V_4 - V_0) + \frac{1}{6}(V_6 - V_2) + \frac{1}{6}(V_5 - V_1) + \frac{1}{3}(V_7 - V_3)$, with analogous expressions for B and χ. The exactness property $\phi_A + \phi_B + \phi_\chi = V_7 - V_0$ holds by construction.



(a) Fertility

(b) Dual Parenthood

(c) Gender Income Gap

(d) CE: Married Female

Figure 9. Shapley-Value Decomposition of Mexico's Transition (1990–2025)

Notes: Each panel plots the total change from the all-off baseline (gray solid) and the three Shapley contributions: A (gender-neutral productivity, navy solid), B (relative wage gap, maroon dashed), and χ (childcare division, green dash-dot). The dashed horizontal line at zero marks the no-change reference. The decomposition is exact in every year. Appendix Figure A.4 reports analogous decompositions for single-female and single-male consumption equivalents.

decline overall, these negative shares indicate offsetting increases. The direction is consistent with Proposition 5: shifting care toward fathers can lower the wife's effective price of children and make marriage more attractive, supporting fertility and dual parenthood as wages converge.

Gender-neutral productivity contributes mainly to demographic change. Rising A accounts for 31 percent of the fertility decline and 95 percent of the dual-parenthood decline through the opportunity cost of children. Its contribution to income convergence is 4 percent, and its contribution to the decline in female consumption equivalents is about 2–3 percent, compared with 22 percent for single men. Since a proportional productivity increase raises both wages, it has a much smaller direct effect on relative earnings than a change in B.

The decomposition connects the policy mechanism to Mexico's longer transition. Changes in

childcare allocation materially support fertility and family structure, even though wage convergence accounts for most of the decline in the income gap. A force can therefore be important for buffering demographic change without accounting for a large share of observed income convergence.

6 Conclusion

This paper studies the joint determination of fertility, dual parenthood, and gender income equality. Across 115 countries, attainment on all three dimensions is substantially less common than statistical independence would imply. The shortfall is similar in OECD and non-OECD countries, but its bottleneck differs: dual parenthood in the OECD, where fertility and gender equality are approximately unrelated, and gender equality elsewhere.

A tractable marriage-market model connects these outcomes through childcare costs and income sharing. The wife chooses fertility conditional on a transfer, and the husband evaluates the resulting consumption–fertility bundle when deciding whether to marry. The transfer adjusts to clear the market. Wage convergence raises the fertility level compatible with income equality but shrinks the income-sharing gains from marriage, which helps explain why the binding outcome differs across country groups.

The policy analysis identifies two routes to joint improvement. Coordinating higher female wages with an appropriately sized marriage benefit improves all three outcomes locally at any interior equilibrium, but requires two instruments to be calibrated together. Redistributing childcare from wives to husbands works through a single margin, provided it lowers the wife's effective childcare cost, leaves room for the husband to gain after the transfer adjusts, and narrows relative earnings after fertility responds. The equilibrium transfer makes this second route bilateral: releasing mothers' time supports marriage only if the revised bundle is also attractive to fathers.

In the Mexico calibration, the model tracks fertility and the broad movement in dual parenthood. A Shapley decomposition attributes most of the demographic and welfare changes to wage convergence, while the more equal division of childcare partly offsets the declines in fertility and dual parenthood. From 2001 onward, all three sufficient conditions hold at the observed allocation. Before then, the large wage gap prevents relative earnings from converging. The paternal share at which the conditions first fail rises from about 24 to 35 percent: above the observed share but below equal sharing in every year.

The broader lesson is that family policy operates through both household choices and participation in the marriage market. A reform's initial effect on one spouse's time or resources can be reinforced or offset by the transfer response and the other spouse's willingness to participate. Accounting for both sides clarifies when progress toward fertility, family stability, and gender income equality can be sustained together.

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Appendix

A Additional Empirical Results

A.1 Binarization Cutoffs and Sample Diagnostics

Table A.1 reports the binarization cutoffs for each sample, alongside summary statistics for the three outcome variables. A country-year is classified as "high" on a dimension if and only if its value strictly exceeds the sample-specific cutoff.

			TFR		Both parents (%)		Female income share	
Sample	Countries	Obs	Median	SD	Median	SD	Median	SD
Full	115	1,764	2.00	1.50	80.1	12.5	0.341	0.079
OECD	39	879	1.56	0.33	83.7	5.6	0.374	0.058
Non-OECD	76	885	3.24	1.52	70.5	14.3	0.294	0.080

Table A.1. Binarization Cutoffs and Sample Statistics

Notes: Cutoffs are the sample-specific medians in original units, displayed with rounding. TFR is children per woman. "Both parents" is the share of children living with both parents (percentage points). Female income share is the female share of national labor income (proportion). SD denotes the standard deviation within each sample. A country-year is classified as "high" on a dimension if its value strictly exceeds the corresponding unrounded cutoff.

Table A.2 reports pairwise correlations in the original variables. All three pairs are negatively correlated in every sample. The strongest full-sample correlation is between fertility and the female income share (-0.52). In the OECD, the fertility–gender equality and dual parenthood–gender equality correlations are moderately negative (-0.42 and -0.31). In non-OECD countries, dual parenthood and gender equality have the strongest correlation (-0.40), while fertility and dual parenthood are only weakly correlated (-0.12). These statistics describe variation in the original outcomes, whereas the main decomposition concerns joint attainment above sample-specific medians. The two measures need not rank associations in the same way.

Sample	TFR–BPP	TFR–FIS	BPP–FIS
Full	-0.41	-0.52	-0.05
OECD	-0.26	-0.42	-0.31
Non-OECD	-0.12	-0.33	-0.40

Table A.2. Pairwise Correlations in Original Units

Notes: TFR is the total fertility rate. BPP is the share of children living with both parents. FIS is the female income share.

A.2 Bootstrap Inference

All bootstrap exercises use 1,000 replications clustered by country. Table A.3 reports detailed bootstrap results for the trinity shortfall. Table A.4 reports bootstrap results for the pairwise contribution decomposition.

	Full		OECD		Non-OECD	
Statistic	Estimate	Boot SE	Estimate	Boot SE	Estimate	Boot SE
Trinity share	0.028	0.014	0.060	0.020	0.061	0.023
Shortfall (pp)	-9.65	1.40	-6.34	2.04	-6.38	2.30
$I(F;M)$ (bits)	0.121	0.054	0.076	0.054	0.014	0.025
$I(F;G)$ (bits)	0.239	0.066	0.000	0.017	0.060	0.044
$I(M;G)$ (bits)	0.015	0.018	0.038	0.037	0.027	0.033
Total correlation (bits)	0.372	0.077	0.119	0.071	0.128	0.056
Interaction information (bits)	0.003	0.029	-0.005	0.023	-0.028	0.026

Table A.3. Bootstrap Inference for Key Statistics

Notes: Bootstrap uses 1,000 replications clustered by country. Shortfall is $p_{111} - p_F p_M p_G$ in percentage points. $I(\cdot;\cdot)$ denotes pairwise mutual information. Total correlation is $H(F) + H(M) + H(G) - H(F, M, G)$. Interaction information is $I(F;M) - I(F;M \mid G)$; negative values indicate synergy, positive values indicate redundancy. Interaction information estimates are not statistically significant at the 5 percent level in any sample.

	OECD				Non-OECD			
Component	Point (pp)	Boot SE	95% CI	p	Point (pp)	Boot SE	95% CI	p
F-M tension	-4.02	1.35	$[-6.63, -1.50]$	0.003	-1.71	1.49	$[-4.29, 1.52]$	0.253
F-G tension	$+0.07$	1.56	$[-3.64, 2.39]$	0.964	-3.57	1.28	$[-5.99, -0.97]$	0.005
M-G tension	-2.86	1.42	$[-5.34, 0.11]$	0.044	-2.42	1.49	$[-4.96, 0.72]$	0.105
Three-way	$+0.47$	1.24	$[-2.07, 2.81]$	0.706	$+1.31$	1.26	$[-0.96, 3.90]$	0.300

Table A.4. Bootstrap Inference for the Pairwise Contribution Decomposition

Notes: Point estimates and bootstrap standard errors are in percentage points. Confidence intervals are percentile-based. The p-value is from a two-sided normal approximation using the point estimate and bootstrap standard error. The F-M tension is significantly negative in the OECD; the F-G tension is significantly negative in the non-OECD. The F-G tension in the OECD is indistinguishable from zero.

The interaction information estimates are negative in the OECD (-0.005 bits) and non-OECD (-0.028 bits) subsamples, suggesting synergy, but neither is statistically significant at the 5 percent level. Rather than emphasizing the sign of interaction information, the paper focuses on the statistically significant trinity shortfall and its decomposition into pairwise tensions, which provides a more directly interpretable characterization of the trade-off.

A.3 Robustness: Gender Wage Gap

The main gender-equality measure combines earnings opportunities with labor-force participation. Fertility can therefore affect the female income share through mothers' market work even when wages remain unchanged. This appendix examines whether the OECD decomposition depends on that direct participation channel.

I replace the WID female income share with the OECD unadjusted gender wage gap, defined as the difference between male and female median full-time earnings as a percentage of male median earnings. The measure compares earnings among employed workers rather than incorporating participation directly. I negate it so that higher values indicate greater equality. The robustness sample includes OECD countries with jointly available wage-gap, dual-parenthood, and TFR data, and I recompute all medians and binary indicators within that sample.

Table A.5 reports an observed joint-attainment share of 8.0 percent, compared with 12.4 percent under independence. The resulting shortfall is 4.43 percentage points, below the main OECD estimate of 6.34 percentage points. The observed-to-independence ratio is 0.64, so joint attainment remains substantially below the benchmark. The comparison changes both the gender-equality measure and the available sample, and therefore serves as a robustness check on the pattern.

Sample	Gender equality measure	Observed (%)	Independence (%)	Shortfall (pp)
OECD (main)	Female income share (WID)	6.03	12.37	−6.34
OECD (robustness)	Neg. gender wage gap (OECD)	8.0	12.4	−4.43

Table A.5. Trinity Shortfall: Main vs. Robustness Gender Equality Measure

Notes: The main OECD sample uses the WID female income share, which incorporates differences in labor-force participation. The robustness sample uses the OECD unadjusted gender wage gap among full-time employed workers (negated so that higher values indicate greater equality), which measures earnings among full-time employed workers rather than incorporating participation directly. Both samples use TFR for fertility and children's co-residence with both parents for dual parenthood. All medians are re-estimated within each sample.

Figure A.1 presents the Venn diagram for the robustness sample. The two largest categories are "F only" (18.4 percent) and "M+G" (17.7 percent), while "F+G" contains 16.5 percent of observations, compared with 19.0 percent in the main OECD sample. With the participation component removed from the gender-equality measure, the pattern resembles a tension between fertility and the other two outcomes, consistent with the decomposition below, in which the fertility–dual parenthood pair accounts for nearly the entire shortfall.

Table A.6 and Figure A.2 compare the decompositions. With the wage-gap measure, the fertility–dual parenthood contribution accounts for virtually the entire shortfall, while the other two pairwise contributions are small. Thus, the prominence of the fertility–dual parenthood association persists when gender equality is measured among full-time employed workers.

Two features are particularly informative. First, the M–G contribution, which accounts for

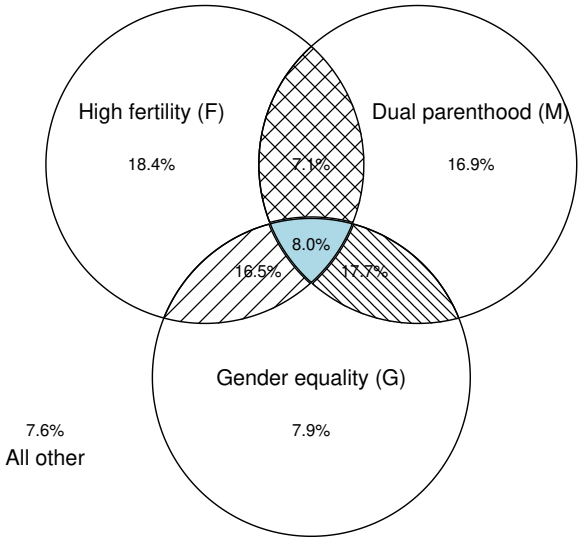


Figure A.1. Three-Way Trade-Off: OECD Robustness Sample (Gender Wage Gap)

Notes: Venn diagram for the OECD robustness sample using the negated gender wage gap among employed workers as the gender equality measure. Each dimension is split at the sample-specific median. Under statistical independence, the central overlap (F+M+G) would contain approximately 12.5 percent of observations.

	OECD (income share)		OECD (wage gap)	
Component	Contribution (pp)	% of shortfall	Contribution (pp)	% of shortfall
F-M tension	−4.02	63.4%	−4.88	≈110%
F-G tension	+0.07	−1.1%	−0.25	≈6%
M-G tension	−2.86	45.1%	+0.42	≈-9%
Three-way residual	+0.47	−7.4%	+0.29	≈-7%
Total shortfall	−6.34	100%	−4.43	100%

Table A.6. Pairwise Contribution Decomposition: Main vs. Robustness Measure

Notes: The left panel reproduces the main OECD decomposition from Table 4. The right panel uses the negated OECD gender wage gap among employed workers. Approximate percentages for the robustness sample are read from the decomposition output; see Figure A.2 for the graphical comparison.

45.1 percent of the shortfall in the main analysis, becomes negligible with the wage-gap measure. This is consistent with participation contributing to the main income-share association, although the comparison does not isolate that channel causally. Second, the F–G contribution remains small under both measures. The defining feature of the OECD bottleneck therefore survives the change in measurement: fertility and gender equality are approximately unrelated, and the shortfall remains concentrated in a pair involving dual parenthood. With the M–G contribution negligible, this sample alone cannot distinguish dual parenthood from fertility as the bottleneck; the main

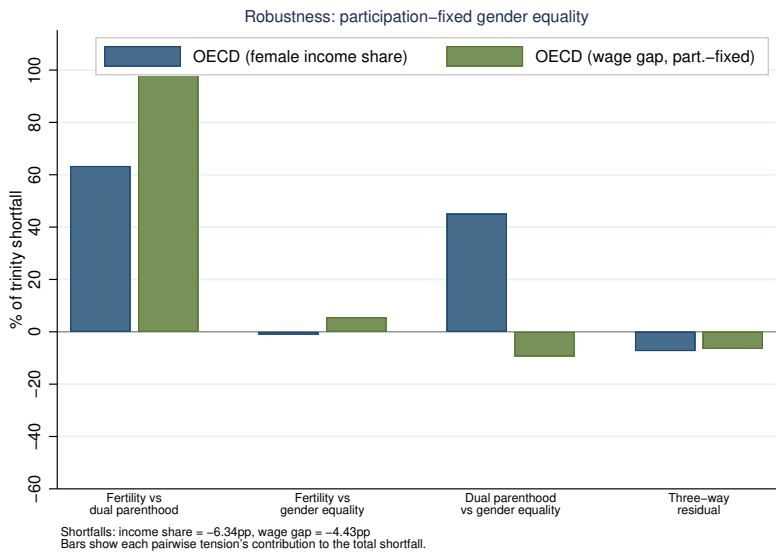


Figure A.2. Decomposition Comparison: Female Income Share vs. Gender Wage Gap

Notes: Bars show each pairwise tension's contribution to the total trinity shortfall, expressed as a percentage. Navy bars use the WID female income share (main analysis); green bars use the negated OECD gender wage gap among employed workers (robustness). The wage-gap measure compares earnings among employed workers and therefore excludes the direct participation component present in the female income share.

income-share analysis, where both pairs involving dual parenthood are significant, provides that distinction.

B Additional Figures

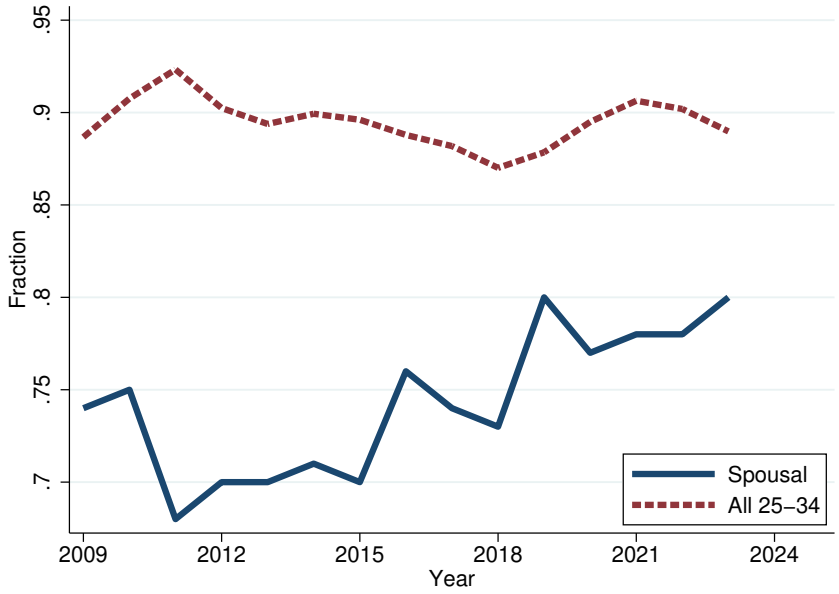
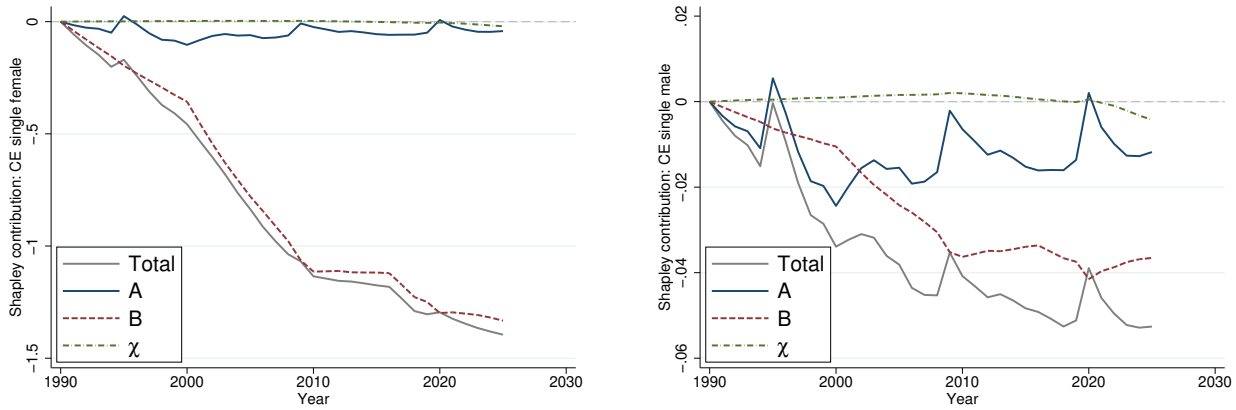


Figure A.3. Gender Earnings Gaps in the U.S.

Notes: The spousal earnings gap uses ACS women aged 18–39, not enrolled in school, childless, and in a first marriage within the past year to a husband also not enrolled in school. The overall gap for full-time wage and salary workers aged 25–34 is from the BLS Annual Highlights of Women's Earnings.



(a) Shapley Decomposition: CE Single Female

(b) Shapley Decomposition: CE Single Male

Figure A.4. Additional Results on Consumption Equivalents

Notes: Each panel plots the total change from the all-off baseline (gray solid) and the three Shapley contributions: A (gender-neutral productivity, navy solid), B (relative wage gap, maroon dashed), and χ (childcare division, green dash-dot). The dashed horizontal line at zero marks the no-change reference. See Table 7 for the Shapley shares at 2025.

C Identification of Calibrated Parameters

The calibration estimates four parameters $(\beta, \rho, \lambda, \theta)$ by matching four moments: mean fertility, the fertility change over the sample, mean dual parenthood, and the dual-parenthood change. This appendix examines local identification by computing the elasticity Jacobian of the moment vector with respect to the parameter vector and its inverse, both evaluated at the calibrated values in Table 6.

Forward Jacobian

Define the elasticity Jacobian $\mathcal{J}$ with entries $\mathcal{J}_{ij} = (\partial m_i / \partial p_j)(p_j / m_i)$, where m_i is the i-th moment and p_j the j-th parameter. The matrix is computed by perturbing each parameter by one percent from its calibrated value and recording the resulting percentage change in each moment. With rows ordered as (mean fertility, fertility change, mean dual parenthood, dual-parenthood change) and columns as $(\beta, \rho, \lambda, \theta)$:

$$\mathcal{J} = \begin{pmatrix} 3.423 & 1.221 & 2.837 & 0.066 \\ 2.292 & 2.905 & -2.984 & 0.154 \\ 0.874 & 0.233 & 3.744 & 0.096 \\ 9.386 & 2.485 & -10.817 & 3.995 \end{pmatrix}. \tag{C.1}$$

For β and λ, the diagonal entry dominates its row (ratios of diagonal to largest off-diagonal entry of 1.21 and 4.28). The fertility change responds about equally to ρ and λ (ratio 0.97). The dual-parenthood change responds more to λ (-10.8) than to θ (4.0; ratio 0.37), because shifting the participation level also changes the gap between the endpoints.

Inverse Jacobian

The inverse $\mathcal{J}^{-1}$ describes the corresponding local sensitivity of parameter estimates to the target moments. Its (i, j) entry gives the percentage change in parameter i associated with a one percent change in moment j, holding the other target moments fixed. Its rows therefore indicate which measurement changes have the largest effects on each estimated parameter:

$$\mathcal{J}^{-1} = \begin{pmatrix} 0.517 & -0.186 & -0.509 & 0.011 \\ -0.415 & 0.492 & 0.628 & -0.027 \\ -0.066 & 0.009 & 0.305 & -0.007 \\ -1.134 & 0.154 & 1.630 & 0.224 \end{pmatrix}. \tag{C.2}$$

Entry (i, j) of $\mathcal{J}^{-1}$ gives the percentage change in parameter i per one percent change in target

j, holding the other targets fixed. β and λ load mainly on the level moments, while ρ loads on several targets. The informative row is θ's. Its intended target, the dual-parenthood change, enters with 0.224, while mean dual parenthood (1.630) and mean fertility (-1.134) have much larger coefficients. The estimate of θ is therefore driven mainly by the level moments, so uncertainty in those moments can translate into large changes in θ. Confidence intervals would additionally require the moments' covariance matrix.

D Proofs for the Baseline Model

D.1 Marriage Probabilities and the Husband's Participation Relation

Let the independent shocks have distribution $F(\varepsilon) = \exp(-\varepsilon^{-\theta})$ for $\varepsilon > 0$. An individual marries if $\varepsilon^m V^{g,m} \geq \varepsilon^s V^{g,s}$. Integrating over $\varepsilon^s = x$ gives

$$\begin{aligned}\Pr(\varepsilon^m V^{g,m} < \varepsilon^s V^{g,s}) &= \int_0^\infty F\left(x\frac{V^{g,s}}{V^{g,m}}\right) dF(x) \\ &= \frac{(V^{g,s})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta}.\end{aligned}$$

Taking the complement yields (3). Strict monotonicity in the utility ratio gives (4).

Single men's value is $(1-\beta)^{1/r} w^{\male}$. Therefore

$$\frac{V^{\male,m}}{V^{\male,s}} = \lambda \left[(1-\alpha)^r + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{\male}}\right)^r\right]^{1/r}. \tag{D.1}$$

Setting this equal to $(\mathcal{M}/(1-\mathcal{M}))^{1/\theta}$ gives the conditional participation schedule

$$\alpha = 1 - \left[\left(\frac{1}{\lambda}\left(\frac{\mathcal{M}}{1-\mathcal{M}}\right)^{1/\theta}\right)^r - \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{\male}}\right)^r\right]^{1/r},$$

where its bracket is positive. At zero fertility this gives $\alpha = 1 - \lambda^{-1}(\mathcal{M}/(1-\mathcal{M}))^{1/\theta}$, which equals zero only when $\mathcal{M}/(1-\mathcal{M}) = \lambda^\theta$. This schedule describes the husband's willingness to participate conditional on fertility; the wife's household problem determines fertility.

D.2 Household Choices and Proof of Lemma 2

Proof. For a woman with budget $c + w^{\female}\chi n = I$, the interior first-order condition is

$$\frac{u_n}{u_c} = \frac{\beta}{1-\beta}\left(\frac{c}{n}\right)^{1/\rho} = w^{\female}\chi.$$

It follows that

$$\frac{c}{n} = \left(\frac{(1-\beta)w^{\female}\chi}{\beta}\right)^\rho,$$

and hence

$$n\left[\left(\frac{(1-\beta)w^{\female}\chi}{\beta}\right)^\rho + w^{\female}\chi\right] = I.$$

Setting $I = w^{♀}$ in singlehood and $I = w^{♀} + \alpha w^{♂}$ in marriage gives (7). Both consumption and fertility scale by $1 + \alpha\Gamma^w$ across states. Homogeneity therefore implies

$$\frac{V^{♀,m}}{V^{♀,s}} = \lambda(1 + \alpha\Gamma^w).$$

At the equilibrium of Lemma 1, $\alpha > 0$, so marital fertility exceeds single fertility. Substitution into (3) proves the marriage-rate expression in (10). □

D.3 Proof of Lemma 1

Proof. Combine (D.1), the wife's utility ratio, and (4). Canceling λ and raising both sides to power r gives

$$(1-\alpha)^r + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{♂}}\right)^r = (1 + \alpha\Gamma^w)^r.$$

Substitute $n^m = (1 + \alpha\Gamma^w)/C$ to obtain

$$(1-\alpha)^r = K(1 + \alpha\Gamma^w)^r.$$

Assumption 1 implies $0 < K < 1$. On $-1/\Gamma^w < \alpha < 1$, taking positive roots is valid:

$$1 - \alpha = K^{1/r}(1 + \alpha\Gamma^w).$$

This linear equation has the unique solution in (9), and substitution gives marital fertility.

The time restriction in (8) guarantees positive married female labor supply. Single female labor supply is positive because $C > \chi$, male consumption is positive because $\alpha < 1$, and female consumption is positive by the first-order condition. CES preferences are strictly quasiconcave on the positive domain, so the feasible interior solutions are unique household optima.

Every interior equilibrium satisfies the preceding equivalent equations. The constructed solution is therefore the unique interior equilibrium. The parameters λ and θ affect participation, while the transfer and fertility formulas depend on the remaining primitives. □

D.4 Aggregate Formulas and the Empirical Family-Structure Mapping

For compactness define

$$g \equiv K^{1/r}, \qquad q \equiv 1 + \alpha^*\Gamma^w = \frac{1 + \Gamma^w}{1 + g\Gamma^w} > 1.$$

Then

$$n^s = \frac{1}{C}, \qquad n^m = \frac{q}{C}, \qquad \mathcal{M} = \frac{(\lambda q)^\theta}{1+(\lambda q)^\theta}, \qquad n = \frac{1+\mathcal{M}(q-1)}{C}. \tag{D.2}$$

The child-weighted empirical counterpart of marriage is

$$\frac{\mathcal{M} n^m}{n} = \frac{\mathcal{M} q}{1+\mathcal{M}(q-1)}.$$

The mapping increases in both marriage and relative marital fertility:

$$\frac{\partial}{\partial \mathcal{M}} \frac{\mathcal{M} q}{1+\mathcal{M}(q-1)} = \frac{q}{[1+\mathcal{M}(q-1)]^2} > 0, \qquad \frac{\partial}{\partial q} \frac{\mathcal{M} q}{1+\mathcal{M}(q-1)} = \frac{\mathcal{M}(1-\mathcal{M})}{[1+\mathcal{M}(q-1)]^2} > 0. \tag{D.3}$$

Marriage and the child-based measure therefore move together whenever $\mathcal{M}$ and q move in the same direction, or one remains unchanged while the other changes strictly. This property connects the individual-policy results to the empirical family-structure measure. Their levels remain distinct because adults and children receive different population weights.

D.5 Proof of Lemma 3

Proof. Part (i) is immediate from (5). Solving $\Gamma^w/(1-\chi n) \le \bar{\Gamma}$ for n gives $\bar{n}(\Gamma^w) = (1/\chi)(1 - \Gamma^w/\bar{\Gamma})$, which is linear and strictly decreasing in Γ^w.

For part (ii), recall from Appendix D.4 that $q - 1 = (1-g)\Gamma^w/(1+g\Gamma^w)$ with $g = K^{1/r}$. The direct partial derivative is

$$\frac{\partial (q-1)}{\partial \Gamma^w} = \frac{1-g}{(1+g\Gamma^w)^2} > 0.$$

Raising Γ^w by lowering $w^{\female}$ at fixed $w^{\male}$ reduces C (since $\rho > 1$), which reduces K and hence g. Because

$$\frac{\partial (q-1)}{\partial g} = -\frac{\Gamma^w(1+\Gamma^w)}{(1+g\Gamma^w)^2} < 0,$$

the indirect channel also raises $q-1$. Marriage $\mathcal{M} = (\lambda q)^\theta/[1+(\lambda q)^\theta]$ is increasing in q, and $\varepsilon_{n,\mathcal{M}} = \mathcal{M}(q-1)/[1+\mathcal{M}(q-1)]$ is increasing in both $\mathcal{M}$ and $q-1$. □

D.6 Proof of Proposition 1

Proof. The objects C, q, and household fertilities are independent of λ. Therefore

$$\mathcal{M}_\lambda = \frac{\theta \mathcal{M}(1-\mathcal{M})}{\lambda} > 0, \qquad n_\lambda = (n^m - n^s)\mathcal{M}_\lambda > 0.$$

With wages and χ fixed, $\Gamma^y = \Gamma^w/(1-\chi n)$ rises. Since q is fixed, (D.3) also shows that the empirical family-structure counterpart rises. □

D.7 Proof of Proposition 2

Proof. At fixed $w^{\male}$, an increase in $w^{\female}$ raises C, K, and g, and lowers Γ^w. Since

$$q_{\Gamma^w} = \frac{1-g}{(1+g\Gamma^w)^2} > 0, \qquad q_g = -\frac{(1+\Gamma^w)\Gamma^w}{(1+g\Gamma^w)^2} < 0,$$

both changes reduce q, so marriage falls. Both $n^s = 1/C$ and $n^m = q/C$ fall. Aggregate fertility satisfies

$$n_{w^{\female}} = \mathcal{M}(n^m)_{w^{\female}} + (1-\mathcal{M})(n^s)_{w^{\female}} + (n^m - n^s)\mathcal{M}_{w^{\female}} < 0.$$

Female wages and female labor supply both rise, so Γ^y falls. The child-based family-structure measure falls because $\mathcal{M}$ and q both fall. These conclusions hold for either sign of $\alpha_{w^{\female}}$. □

D.8 Proof of Proposition 3

Proof. Reducing χ lowers C, K, and g at fixed Γ^w, so q and marriage rise. Household fertilities rise, and the shift toward marriage reinforces aggregate fertility.

To determine the income response, write

$$\chi n = \frac{\chi}{C}[1 + \mathcal{M}(q-1)], \qquad \frac{\chi}{C} = \frac{1}{1 + \left(\frac{1-\beta}{\beta}\right)^{\rho}(w^{\female}\chi)^{\rho-1}}.$$

Since $\rho > 1$, the first factor rises as χ falls. The second also rises because $\mathcal{M}$ and q increase and $q > 1$. Hence female labor supply falls and the income gap widens. The empirical family-structure counterpart rises by (D.3). □

D.9 Conditional Feasibility and Proof of Proposition 4

Derivation of (11). Equation (10) gives

$$\frac{1}{\lambda}\left(\frac{\mathcal{M}}{1-\mathcal{M}}\right)^{1/\theta} = 1 + \alpha^*\Gamma^w.$$

Since $\alpha^* < 1$,

$$1 + \alpha^*\Gamma^w < 1 + \Gamma^w = 1 + \Gamma^y(1-\chi n).$$

This proves the necessary restriction. □

Proof of Proposition 4. Hold $w^{\male}$ and χ fixed. Start with a female-wage increase and choose the marriage-benefit adjustment to keep aggregate fertility locally constant:

$$\frac{d\lambda}{dw^{\female}} = -\frac{n_{w^{\female}}}{n_\lambda} > 0.$$

The implicit function theorem applies because $n_\lambda > 0$. Household fertilities are independent of λ, so along this path

$$0 = \mathcal{M}(n^m)_{w^{\female}} + (1-\mathcal{M})(n^s)_{w^{\female}} + (n^m - n^s)\frac{d\mathcal{M}}{dw^{\female}}.$$

The first two terms are negative and $n^m > n^s$, implying $d\mathcal{M}/dw^{\female} > 0$. The income gap falls because the female wage rises while female labor supply $l^{\female} = 1 - \chi n$ remains constant.

Choose a slightly larger positive slope for λ. Aggregate fertility then rises strictly. Continuity preserves the strict marriage increase and income-gap decline for a sufficiently small change. Thus all three objectives improve.

The child-based measure improves as well. On the constant-fertility path it equals $1-(1-\mathcal{M})n^s/n$, which rises because marriage increases and single fertility falls. This strict increase persists under the sufficiently small additional marriage-benefit adjustment. □

E Extended Model with Gender-Differentiated Childcare

This appendix derives the household choices, equilibrium responses, and redistribution results used in Section 4. Throughout, the father's raw time requirement per child is $\chi^{\male}/\eta$, where $\eta \in (0,1]$ measures paternal care efficiency. Setting $\eta = 1$ recovers the equal-efficiency case.

E.1 Household Solutions and Proof of Lemma 4

Write the wife's monetary price of fertility as

$$P \equiv \tilde{w}^{\female}\tilde{\chi}^{\female} = w^{\female}\chi^{\female} + \alpha w^{\male}\frac{\chi^{\male}}{\eta}.$$

Throughout this appendix, household allocations are interior, $P > 0$, and both spouses have positive consumption and market labor supply. We retain z and R as defined in the main text.

Proof. The wife's budget is

$$c^{\female,m} + Pn^m = \tilde{w}^{\female}.$$

Her first-order condition gives

$$\frac{c^{♀,m}}{n^m} = \left(\frac{(1-\beta)P}{\beta}\right)^{\rho}.$$

Consequently,

$$\begin{aligned} n^m &= \frac{\tilde{w}^{♀}}{P + \left(\frac{(1-\beta)P}{\beta}\right)^{\rho}}, \\ c^{♀,m} &= \frac{\tilde{w}^{♀}}{1 + \left(\frac{\beta}{1-\beta}\right)^{\rho} P^{1-\rho}}. \end{aligned} \tag{E.1}$$

Using $P = \tilde{w}^{♀}\tilde{\chi}^{♀}$ in the fertility expression proves (14). Substituting the optimal allocation into CES utility gives

$$V^{♀,m} = \lambda(1-\beta)^{1/r}\tilde{w}^{♀}\left[1 + \left(\frac{\beta}{1-\beta}\right)^{\rho} P^{1-\rho}\right]^{1/(\rho-1)},$$

which proves (15). The singlehood formula follows by setting $\lambda = 1$, $\tilde{w}^{♀} = w^{♀}$, and $P = w^{♀}\chi$.

At fixed $\tilde{w}^{♀}$, fertility and indirect utility decrease with P. The consumption expression in (E.1) increases with P because $\rho > 1$. A lower fertility price therefore raises fertility and utility while lowering consumption. □

E.2 Equilibrium Derivatives

Partial derivatives with respect to δ hold α fixed; primes denote total equilibrium derivatives. All expressions are evaluated at the initial equilibrium. At $\chi^{♂} = 0$, policy derivatives are interpreted as right-hand derivatives.

Write the spouses' log marriage premiums as

$$f \equiv \log \frac{V^{♀,m}}{V^{♀,s}}, \qquad h \equiv \log \frac{V^{♂,m}}{V^{♂,s}}.$$

Market clearing requires $f = h$. Recall that

$$z = \frac{Pn^m}{\tilde{w}^{♀}} \in (0,1).$$

Differentiating the household fertility formula gives

$$\begin{aligned} n^m_\alpha &= n^m w^{\male} \left[\frac{1}{\tilde{w}^{\female}} - \frac{\chi^{\male}/\eta}{P}\big(\rho - (\rho-1)z\big) \right], \\ n^m_\delta &= \frac{n^m(w^{\female} - \alpha w^{\male}/\eta)}{P}\big(\rho - (\rho-1)z\big). \end{aligned} \tag{E.2}$$

Here $P_\alpha = w^{\male}\chi^{\male}/\eta$ and $P_\delta = -(w^{\female} - \alpha w^{\male}/\eta)$.

The wife's envelope condition yields

$$f_\alpha = \frac{w^{\male}(1 - \frac{\chi^{\male}}{\eta}\, n^m)}{\tilde{w}^{\female}} > 0, \qquad f_\delta = \frac{(w^{\female} - \alpha w^{\male}/\eta)\, n^m}{\tilde{w}^{\female}}. \tag{E.3}$$

To see this, homogeneity and the wife's first-order condition imply

$$\frac{u_c(c^{\female,m}, n^m)}{u(c^{\female,m}, n^m)} = \frac{1}{\tilde{w}^{\female}}.$$

For the husband, Euler's theorem and the definition of R imply

$$c^{\male,m} + \frac{u_n(c^{\male,m}, n^m)}{u_c(c^{\male,m}, n^m)} n^m = (1-\alpha)w^{\male} + Rn^m > 0.$$

His log-premium derivatives are therefore

$$\begin{aligned} h_\alpha &= \frac{-w^{\male}(1 - \frac{\chi^{\male}}{\eta}\, n^m) + Rn^m_\alpha}{(1-\alpha)w^{\male} + Rn^m}, \\ h_\delta &= \frac{-(1-\alpha)w^{\male}\, n^m/\eta + Rn^m_\delta}{(1-\alpha)w^{\male} + Rn^m}. \end{aligned} \tag{E.4}$$

Singlehood values remain fixed throughout this experiment.

Lemma (Local equilibrium responses)**.** *At an interior equilibrium with $f_\alpha - h_\alpha \neq 0$, there is a differentiable local equilibrium branch. Writing $G \equiv f' = h'$ for the common log-premium*

response,

$$G = \frac{f_\alpha h_\delta - h_\alpha f_\delta}{f_\alpha - h_\alpha}, \tag{E.5}$$

$$\mathcal{M}' = \theta \mathcal{M}(1-\mathcal{M})G, \tag{E.6}$$

$$\alpha' = \frac{\tilde{w}^{\female} G - (w^{\female} - \alpha w^{\male}/\eta)\, n^m}{w^{\male}(1 - \frac{\chi^{\male}}{\eta}\, n^m)}, \tag{E.7}$$

$$(n^m)' = n^m_\delta + n^m_\alpha \alpha'. \tag{E.8}$$

The responses also satisfy

$$R(n^m)' = \big(w^{\male} + w^{\female} + R n^m\big)G + \big(w^{\male}/\eta - w^{\female}\big)n^m. \tag{E.9}$$

Proof. Apply the implicit function theorem to $f - h = 0$. Differentiating market clearing gives

$$\alpha' = \frac{h_\delta - f_\delta}{f_\alpha - h_\alpha}.$$

Substitution into $G = f_\delta + f_\alpha \alpha'$ proves (E.5). Differentiating the participation rule proves (E.6). Solving the wife's envelope equation for α' gives (E.7), and the chain rule gives (E.8).

To obtain (E.9), write the spouses' total log-premium differentials as

$$\tilde{w}^{\female}\, G = w^{\male}\big(1 - \tfrac{\chi^{\male}}{\eta}\, n^m\big)\alpha' + \big(w^{\female} - \alpha w^{\male}/\eta\big)n^m$$

and

$$\big((1-\alpha)w^{\male} + R n^m\big)G = -w^{\male}\big(1 - \tfrac{\chi^{\male}}{\eta}\, n^m\big)\alpha' - (1-\alpha)w^{\male}\, n^m/\eta + R(n^m)'.$$

Adding these equations eliminates the transfer response. Since $\tilde{w}^{\female} + (1-\alpha)w^{\male} = w^{\female} + w^{\male}$, and $(w^{\female} - \alpha w^{\male}/\eta) - (1-\alpha)w^{\male}/\eta = w^{\female} - w^{\male}/\eta$, the result is (E.9). □

Single fertility is fixed, so aggregate responses satisfy

$$n' = \mathcal{M}(n^m)' + \mathcal{M}'(n^m - n^s), \tag{E.10}$$

and

$$(\mathcal{M} n^m)' = \mathcal{M}(n^m)' + \mathcal{M}' n^m. \tag{E.11}$$

The child-based family-structure response is

$$\left(\frac{\mathcal{M}n^m}{n}\right)' = \frac{n^s\left[n^m\mathcal{M}' + \mathcal{M}(1-\mathcal{M})(n^m)'\right]}{n^2}.$$

Thus rising marriage and rising marital fertility imply an increase in the child-based counterpart.

E.3 The Male-Side Compensated Interpretation

Consider an auxiliary transfer adjustment that holds the wife's log marriage premium fixed. Denote this adjustment by α_c'. The wife's envelope condition gives

$$0 = f_\delta + f_\alpha \alpha_c', \qquad \alpha_c' = -\frac{(w^{♀} - \alpha w^{♂}/\eta)\, n^m}{w^{♂}\left(1 - \frac{\chi^{♂}}{\eta}\, n^m\right)}.$$

Substituting into (E.2) gives the compensated fertility response:

$$\frac{(n^m)_c'}{n^m} = \frac{(w^{♀} - \alpha w^{♂}/\eta)\,\rho(1-z)}{P\left(1 - \frac{\chi^{♂}}{\eta}\, n^m\right)}.$$

The husband's compensated log-premium derivative is

$$\begin{aligned} h_c' &= h_\delta + h_\alpha \alpha_c' \\ &= \frac{n^m}{\left(1 - \frac{\chi^{♂}}{\eta}\, n^m\right)\left((1-\alpha)w^{♂} + Rn^m\right)} \\ &\quad \times \left[\frac{R(w^{♀} - \alpha w^{♂}/\eta)\,\rho(1-z)}{P} - \left(1 - \frac{\chi^{♂}}{\eta}\, n^m\right)\left(w^{♂}/\eta - w^{♀}\right)\right]. \end{aligned} \tag{E.12}$$

Its denominator is positive. Since $P = \tilde{w}^{♀}\tilde{\chi}^{♀}$, Condition 3 is exactly the requirement $h_c' > 0$.

This auxiliary path compensates the wife rather than clearing the marriage market. To connect it to equilibrium, use (E.5) to obtain

$$G = \frac{f_\alpha}{f_\alpha - h_\alpha} h_c'.$$

The proof below establishes $f_\alpha - h_\alpha > 0$ under Conditions 1 and 3. A positive compensated gain for the husband then raises the common equilibrium marriage premium.

E.4 Proof of Proposition 5

Proof. Condition 1 gives $w^{\female} - \alpha w^{\male}/\eta > 0$. Together with Condition 3, $w^{\male} > w^{\female}$, and interiority, this implies $R > 0$.

Using (E.2)–(E.4),

$$f_\alpha - h_\alpha = \frac{w^{\male}}{\tilde{w}^{\female}\big((1-\alpha)w^{\male} + Rn^m\big)} \times \left[(w^{\male} + w^{\female})\big(1 - \tfrac{\chi^{\male}}{\eta}\, n^m\big) + \frac{R\,\frac{\chi^{\male}}{\eta}\, n^m\, \tilde{w}^{\female}\, \rho(1-z)}{P}\right] > 0. \tag{E.13}$$

Thus the equilibrium is locally regular.

Condition 3 implies $h_c' > 0$. Since $f_\alpha > 0$ and $f_\alpha - h_\alpha > 0$, the compensated comparison gives

$$G = \frac{f_\alpha}{f_\alpha - h_\alpha} h_c' > 0.$$

Equation (E.6) therefore gives $\mathcal{M}' > 0$. Singlehood values are fixed, so both spouses' marital values and marriage-to-singlehood utility ratios rise.

Marital fertility rises directly from (E.9):

$$R(n^m)' = \big(w^{\male} + w^{\female} + Rn^m\big)G + \big(w^{\male}/\eta - w^{\female}\big)n^m > 0.$$

Because $R > 0$, this implies $(n^m)' > 0$.

Under fixed total care,

$$P = w^{\female}\chi - (w^{\female} - \alpha w^{\male}/\eta)\chi^{\male} \le w^{\female}\chi, \qquad \tilde{w}^{\female} > w^{\female}.$$

Fertility increases with resources and decreases with its monetary price. Comparing the married and single household problems therefore gives $n^m > n^s$. Equation (E.10) now implies $n' > 0$.

Differentiating aggregate labor supplies gives

$$(l^{\male})' = -\left[\frac{1}{\eta}\,\mathcal{M}n^m + \frac{\chi^{\male}}{\eta}\,(\mathcal{M}n^m)'\right] < 0$$

and

$$(l^{\female})' = \mathcal{M}n^m + \chi^{\male}(\mathcal{M}n^m)' - \chi n'.$$

At fixed wages,

$$\begin{aligned}\frac{d\log\Gamma^{y}}{d\delta} &= \frac{(l^{\male})'}{l^{\male}} - \frac{(l^{\female})'}{l^{\female}} \\ &= \frac{\chi n'}{l^{\female}} - \left[\mathcal{M}n^{m} + \chi^{\male}(\mathcal{M}n^{m})'\right]\left(\frac{1}{\eta\, l^{\male}} + \frac{1}{l^{\female}}\right).\end{aligned} \tag{E.14}$$

Condition 2 is exactly the requirement that this expression be negative. It does not require female labor supply to rise.

The strict derivative signs and interiority imply the same directional changes for sufficiently small feasible reforms. The child-based family-structure counterpart also rises, because marriage and marital fertility both increase. □

E.5 Redistribution Starting from the Baseline

At $\chi^{\male} = 0$ and $\chi^{\female} = \chi$,

$$P = w^{\female}\chi, \qquad R = \frac{\beta}{1-\beta}\left(\frac{(1-\alpha)w^{\male}}{n^{m}}\right)^{1/\rho}, \qquad z = \frac{w^{\female}\chi n^{m}}{w^{\female} + \alpha w^{\male}}.$$

Substitution into the equilibrium derivatives gives

$$G = \frac{n^{m}}{w^{\male} + w^{\female}}\left[\frac{R(w^{\female} - \alpha w^{\male}/\eta)\,\rho(1-z)}{w^{\female}\chi} - \left(w^{\male}/\eta - w^{\female}\right)\right], \tag{E.15}$$

and

$$\frac{(n^{m})'}{n^{m}} = G + \frac{(w^{\female} - \alpha w^{\male}/\eta)\,\rho(1-z)}{w^{\female}\chi}. \tag{E.16}$$

The male-side condition becomes

$$R(w^{\female} - \alpha w^{\male}/\eta)\,\rho(1-z) > w^{\female}\chi\left(w^{\male}/\eta - w^{\female}\right).$$

Since $l^{\male} = 1$ initially, the income gap narrows if and only if

$$\chi n' < \frac{\mathcal{M}n^{m}}{\eta}\left(\eta + l^{\female}\right).$$

Marriage and aggregate fertility derivatives follow from (E.6) and (E.10).

E.6 Different Total Childcare Requirements by Marital Status

The household and equilibrium derivative formulas remain valid if $\chi^{♀} + \chi^{♂} \neq \chi$, provided both total requirements remain fixed during redistribution and household choices remain interior.

The aggregate fertility conclusion additionally requires $n^m \geq n^s$. Together with increasing marriage and strictly increasing marital fertility, this is sufficient for $n' > 0$ by (E.10).

Labor supplies are

$$l^{♂} = 1 - \frac{\chi^{♂}}{\eta}\mathcal{M}n^m, \qquad l^{♀} = 1 - \chi^{♀}\mathcal{M}n^m - \chi(1-\mathcal{M})n^s.$$

Their derivatives are

$$(l^{♂})' = -\frac{1}{\eta}\mathcal{M}n^m - \frac{\chi^{♂}}{\eta}(\mathcal{M}n^m)',$$

and

$$(l^{♀})' = \mathcal{M}n^m - \chi^{♀}(\mathcal{M}n^m)' + \chi n^s\mathcal{M}'.$$

The exact condition for a declining income ratio is

$$\frac{\mathcal{M}n^m - \chi^{♀}(\mathcal{M}n^m)' + \chi n^s\mathcal{M}'}{l^{♀}} + \frac{\frac{1}{\eta}\mathcal{M}n^m + \frac{\chi^{♂}}{\eta}(\mathcal{M}n^m)'}{l^{♂}} > 0.$$

The child-based family-structure counterpart still rises whenever marriage and marital fertility rise, because single fertility remains fixed.

E.7 Proof of Proposition 6

Proof. Recall that

$$P = w^{♀}\chi^{♀} + \alpha w^{♂}\frac{\chi^{♂}}{\eta}, \qquad r = \frac{\rho - 1}{\rho} \in (0,1).$$

Consider an interior equilibrium with $\chi^{♂} > 0$ satisfying Condition 1. We first derive a necessary restriction for $R \geq 0$.

Marriage-market clearing. The wife's full resources exceed those available in singlehood, $\tilde{w}^{♀} = w^{♀} + \alpha w^{♂} > w^{♀}$, while her monetary price of fertility satisfies

$$P = w^{♀}\chi - \left(w^{♀} - \frac{\alpha w^{♂}}{\eta}\right)\chi^{♂} < w^{♀}\chi.$$

The interior indirect utility formula in Appendix E.1 is increasing in full resources and decreasing in the fertility price. Thus $u(c^{♀,m}, n^m) > V^{♀,s}$. Canceling the common marriage benefit λ from (4) gives

$$\frac{u(c^{♂,m}, n^m)}{u(w^♂, 0)} = \frac{u(c^{♀,m}, n^m)}{V^{♀,s}} > 1.$$

Since $u(w^♂, 0) = (1-\beta)^{1/r} w^♂$, it follows that

$$\frac{u(c^{♂,m}, n^m)^r}{1-\beta} > (w^♂)^r. \tag{E.17}$$

This comparison holds regardless of whether λ is above or below one.

The husband's marginal value of fertility. The wife's first-order condition is

$$\frac{\beta}{1-\beta}\left(\frac{c^{♀,m}}{n^m}\right)^{1/\rho} = P.$$

Hence the husband's net marginal value of fertility can be written as

$$R = P\left(\frac{c^{♂,m}}{c^{♀,m}}\right)^{1/\rho} - (1-\alpha)w^♂\frac{\chi^♂}{\eta}.$$

Suppose $R \geq 0$. All consumption levels and the father's raw childcare requirement are strictly positive, so

$$(c^{♂,m})^{-1/\rho} \leq \frac{P}{(1-\alpha)w^♂\chi^♂/\eta}(c^{♀,m})^{-1/\rho}.$$

Multiplying by the husband's consumption,

$$c^{♂,m} = (1-\alpha)w^♂\left(1 - \frac{\chi^♂}{\eta}n^m\right),$$

yields

$$(c^{♂,m})^r \leq \frac{P}{\chi^♂/\eta}\left(1 - \frac{\chi^♂}{\eta}n^m\right)(c^{♀,m})^{-1/\rho}.$$

The wife's first-order condition also implies

$$\frac{\beta}{1-\beta}(n^m)^r = Pn^m(c^{♀,m})^{-1/\rho}.$$

Adding these expressions gives

$$\frac{u(c^{\male,m},n^m)^r}{1-\beta} = (c^{\male,m})^r + \frac{\beta}{1-\beta}(n^m)^r \\ \leq \frac{P}{\chi^{\male}/\eta}(c^{\female,m})^{-1/\rho}. \tag{E.18}$$

Combining (E.17) and (E.18), and using $r\rho = \rho - 1$, gives

$$c^{\female,m} < \frac{\left[P/(\chi^{\male}/\eta)\right]^{\rho}}{(w^{\male})^{\rho-1}}. \tag{E.19}$$

A restriction on the childcare share. Condition 1 implies $\alpha w^{\male} < \eta w^{\female}$. Therefore

$$\frac{P}{\chi^{\male}/\eta} = \eta w^{\female}\frac{\chi - \chi^{\male}}{\chi^{\male}} + \alpha w^{\male} < \frac{\eta w^{\female}\chi}{\chi^{\male}}. \tag{E.20}$$

Positive paternal market labor supply requires $n^m < \eta/\chi^{\male}$. The wife's consumption consequently satisfies

$$c^{\female,m} = w^{\female}\left[1-(\chi-\chi^{\male})n^m\right] + \alpha w^{\male}\left(1-\frac{\chi^{\male}}{\eta}n^m\right) \\ > w^{\female}\left(1+\eta-\frac{\eta\chi}{\chi^{\male}}\right). \tag{E.21}$$

The strict inequality follows from the positive transfer and remains valid when $\chi^{\male} = \chi$. Combining (E.19)–(E.21) yields

$$1+\eta-\frac{\eta\chi}{\chi^{\male}} < (\Gamma^w)^{1-\rho}\left(\frac{\eta\chi}{\chi^{\male}}\right)^{\rho}.$$

Multiplying by $(\chi^{\male}/\chi)^{\rho} > 0$, we obtain

$$\left[(1+\eta)\frac{\chi^{\male}}{\chi} - \eta\right]\left(\frac{\chi^{\male}}{\chi}\right)^{\rho-1} < \eta^{\rho}(\Gamma^w)^{1-\rho}. \tag{E.22}$$

Thus Condition 1 and $R \geq 0$ jointly require (E.22).

Existence of the ceiling. The left side of (17), viewed as a function of $\bar{\omega}$, equals zero at $\eta/(1+\eta)$ and one at 1. Its derivative is

$$\bar{\omega}^{\rho-2}\left[\rho(1+\eta)\bar{\omega}-(\rho-1)\eta\right]>0 \qquad \text{for } \bar{\omega}\in\left[\frac{\eta}{1+\eta},1\right].$$

Moreover,

$$0<\eta^{\rho}(\Gamma^{w})^{1-\rho}<1,$$

because $\rho>1$, $\Gamma^{w}>1$, and $\eta\in(0,1]$. Equation (17) therefore has a unique solution $\bar{\omega}\in(\eta/(1+\eta),1)$. Whenever $\chi^{\male}/\chi\geq\bar{\omega}$, inequality (E.22) fails. At such an equilibrium, Condition 1 must therefore imply $R<0$.

Finally, under Condition 1, the coefficient multiplying R on the left side of Condition 3 is positive. Its right side is also positive, since

$$1-\frac{\chi^{\male}}{\eta}n^{m}>0, \qquad \frac{w^{\male}}{\eta}-w^{\female}>0.$$

Conditions 1 and 3 thus require $R>0$, which is impossible at or above the ceiling. The case $\chi^{\male}=0$ already lies strictly below $\bar{\omega}$. This completes the proof. □

E.8 An Exact Local Characterization of Joint Improvement

Proposition 5 gives economically interpretable sufficient conditions for joint improvement. The equilibrium derivatives also provide an exact local criterion without imposing Conditions 1 and 3. Maintain the shared-childcare model and the fixed-total-care experiment in (12), with wages, preferences, and η held fixed. Consider an interior equilibrium with $w^{\male}>w^{\female}>0$, $0<\alpha<1$, $\Gamma^{y}>1$, and $\chi^{\female}>0$, so further redistribution toward fathers is feasible. Assume $f_{\alpha}-h_{\alpha}\neq0$, ensuring a differentiable local equilibrium branch. At $\chi^{\male}=0$, derivatives are interpreted as right-hand derivatives.

Retain G from (E.5) and write the equilibrium marital-fertility response as

$$\nu\equiv(n^{m})'=n^{m}_{\delta}+n^{m}_{\alpha}\frac{h_{\delta}-f_{\delta}}{f_{\alpha}-h_{\alpha}}. \tag{E.23}$$

The partial derivatives in this expression are given in (E.2)–(E.4). Neither G nor ν requires the sufficient conditions to hold.

Proposition (Exact local improvement criterion)**.** *Under the preceding assumptions, strict first-order joint improvement in aggregate fertility, adult marriage, and gender income equality occurs*

if and only if Condition 2 holds and

$$G > 0, \qquad \nu + \theta(1 - \mathcal{M})(n^m - n^s)G > 0. \tag{E.24}$$

If family structure is instead measured by children's dual parenthood, $\mathcal{M}n^m/n$, replace $G > 0$ with

$$\nu + \theta n^m G > 0. \tag{E.25}$$

All derivatives in Condition 2 are evaluated along the same equilibrium branch.

Proof. Equations (E.6), (E.10), and (E.11) give

$$\begin{aligned} \mathcal{M}' &= \theta \mathcal{M}(1 - \mathcal{M})G, \\ n' &= \mathcal{M}\left[\nu + \theta(1 - \mathcal{M})(n^m - n^s)G\right], \\ (\mathcal{M}n^m)' &= \mathcal{M}\left[\nu + \theta(1 - \mathcal{M})n^m G\right]. \end{aligned}$$

Since $0 < \mathcal{M} < 1$, the inequalities in (E.24) are equivalent to $\mathcal{M}' > 0$ and $n' > 0$. The accounting identity (E.14) holds independently of the sufficient conditions and makes Condition 2 equivalent to $(\Gamma^y)' < 0$. With $\Gamma^y > 1$, this represents greater income equality locally. Finally,

$$\left(\frac{\mathcal{M}n^m}{n}\right)' = \frac{n^s \mathcal{M}(1 - \mathcal{M})}{n^2}\left[\nu + \theta n^m G\right],$$

whose sign is determined by (E.25). □

The fertility inequality separates changes within marriages from changes in family composition. The response ν includes both the direct care reallocation and the equilibrium transfer adjustment. When $n^m > n^s$, increasing marriage contributes positively to aggregate fertility; the exact condition requires any decline in marital fertility to be smaller than this composition gain. Condition 1, by contrast, restricts the wife's effective childcare price at the initial transfer and helps establish the stronger fertility conclusions in Proposition 5.

The other two conditions have a direct connection to the exact criterion. Condition 2 is already necessary and sufficient for the income ratio to decline. Appendix E.3 gives $G = f_\alpha h_c'/(f_\alpha - h_\alpha)$, with $f_\alpha > 0$ and Condition 3 equivalent to $h_c' > 0$. On a branch with $f_\alpha - h_\alpha > 0$, that condition is therefore necessary and sufficient for increasing adult marriage. On such a branch, the exact test combines Conditions 3 and 2 with the aggregate-fertility inequality in (E.24); unlike Proposition 5, it maintains rather than derives the positive-slope restriction.

These strict first-order tests guarantee improvement for sufficiently small feasible reforms. A zero derivative requires higher-order analysis. The characterization is local to the evaluated branch:

a first failure does not exclude improvement at another allocation or gains from a finite reform relative to the original allocation.

F Dynamic Extension: Cross-Generational Feedback

The static model takes the gender wage gap as given. This appendix allows current family structure to affect that gap in the next generation, creating a feedback from today's marriage market to tomorrow's earning opportunities.

A substantial literature links children's outcomes to their family environment and finds that boys are often more sensitive than girls to disadvantage and the absence of a biological father (Sommers, 2001; Bertrand and Pan, 2013; Chetty et al., 2016; Autor et al., 2019; Wasserman, 2020; Reeves, 2022; Frimmel et al., 2024). Autor et al. (2019) examine gender differences associated with family disadvantage, while Autor et al. (2023) study family structure and high-school performance.

This evidence motivates a reduced-form connection between current marriage and future male relative human capital. If lower marriage weakens boys' subsequent earning opportunities relative to girls', it also changes the resources supporting the next generation's marital contracts.

I introduce a reduced-form law of motion:

$$\Gamma^{w}_{t+1} = \mathcal{H}(\mathcal{M}_t), \qquad \mathcal{H}'(\mathcal{M}_t) > 0. \tag{F.1}$$

Marriage serves as a summary predictor of the family environment. Equation (F.1) captures boys' greater sensitivity to that environment, allowing the underlying mechanism to operate through parental inputs, role models, or peer and neighborhood effects. The relationship is imposed as a modeling assumption; establishing its causal magnitude and its connection to children's living arrangements would require separate empirical analysis.

Figure A.5 brings the channels together. The human-capital link runs across generations; time subscripts are omitted from the diagram to keep the mechanism visible.

The transition unfolds through two exogenous drivers and one endogenous feedback:

(1) **Gender-neutral productivity growth** raises the opportunity cost of children. Fertility falls within both marital states, and the equilibrium transfer and income-sharing multiplier decline. Marriage becomes less attractive, so the shift toward singlehood reinforces lower aggregate fertility. Women devote less time to childcare, and the income gap narrows.

(2) **Declining gender-biased productivity** improves women's wages at a given male wage. Children become more costly in forgone female earnings, while the transfer adds less relative to women's own resources. Fertility and marriage fall, and the income gap narrows through wages and market work.

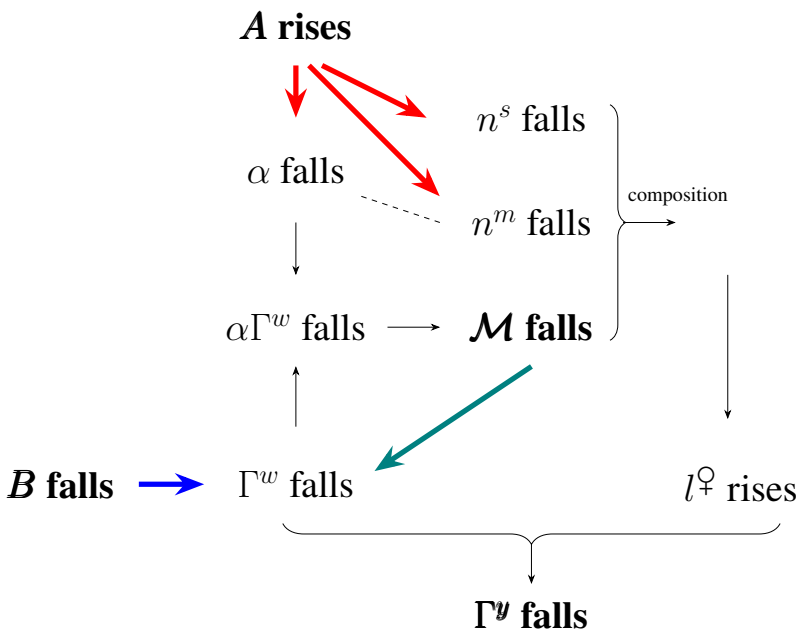


Figure A.5. The Dynamic Mechanism

Notes: Red arrows represent the effects of rising gender-neutral productivity A_t. The blue arrow represents the effect of falling gender-biased productivity B_t. The teal arrow represents the dynamic human-capital effect through $\mathcal{M}_t \to \Gamma^w_{t+1}$ in Equation (F.1). Time subscripts are suppressed. The diagram summarizes directions of propagation; their magnitudes and the stability of the system depend on the response elasticities.

(3) **Dynamic propagation** carries current family structure into future earning opportunities. Lower marriage reduces the next generation's male relative human capital through (F.1). The resulting wage convergence weakens the income-sharing attraction of marriage and puts further downward pressure on fertility. Lemma 3 also links that convergence to a higher fertility ceiling compatible with income equality and a weaker marital-fertility composition channel.

The sign restriction in (F.1) determines the direction of this feedback but leaves its strength open. A steep $\mathcal{H}(\cdot)$ combined with responsive marriage participation could generate persistent amplification, while a flatter relationship or a weaker participation response would allow the effect to decay. Distinguishing these cases requires specifying and estimating the human-capital law of motion using data that connect family structure to subsequent earning opportunities.

The extension has two qualitative implications. First, technological change can affect fertility, marriage, and income inequality together through the same equilibrium system, complementing the emphasis on factor-biased technical change in Greenwood et al. (2005b) and Ngai and Petrongolo (2017). Second, policies that change marriage today may alter the earning opportunities that support future marriages. Their long-run effects depend on the human-capital response as well as the immediate demographic adjustment.

G Robustness of the Quantitative Results to the Childcare Time Cost

The baseline sets the childcare time cost at $\chi = 0.075$, following De La Croix and Doepke (2003). Because this parameter is not estimated from Mexican microdata, I repeat the calibration for $\chi \in \{0.05,\ 0.10\}$, bracketing the baseline. For each value, I reconstruct wages from the accounting identities, re-estimate $(\beta, \rho, \lambda, \theta)$ to match fertility and dual parenthood, and solve the equilibrium in every year. GDP per capita, female income shares, and the ENUT-based childcare division remain unchanged. The exercise examines the sensitivity of model fit, welfare, the paternal-efficiency boundary, and the fertility decomposition.

Table A.7 reports the re-estimated parameters for each specification alongside the baseline.

χ	β	ρ	λ	θ
0.050	0.044	3.0	0.965	24
0.075 (baseline)	0.068	3.0	0.945	18
0.100	0.093	3.0	0.920	16

Table A.7. Re-Estimated Parameters Under Alternative χ

Notes: For each value of χ, the four parameters $(\beta, \rho, \lambda, \theta)$ are re-estimated to match Mexico's fertility and dual-parenthood series over 1990–2025. The middle row reproduces Table 6.

The joint re-estimation accommodates the alternative childcare costs through changes in preferences and marriage parameters. At $\chi = 0.05$, the fertility weight β falls from 0.068 to 0.044, offsetting the higher fertility that a lower time requirement would otherwise encourage. The common marriage benefit λ rises from 0.945 to 0.965, and participation responsiveness θ increases from 18 to 24. At $\chi = 0.10$, β rises to 0.093, λ falls to 0.920, and θ decreases from 18 to 16. Because wages and parameters are jointly recalibrated, these movements describe how the model preserves the targets under each specification. All three estimates remain in the substitution-dominant region ($\rho > 1$).

Figure A.6 reports the results across six panels. Three findings emerge.

Model fit is preserved. Panels (a) and (b) overlay the data with model predictions under each χ. All three specifications track the observed fertility decline and the level and trend of dual parenthood. The fit differences across specifications are modest relative to the overall magnitude of the transitions.

The gender income gap and welfare trajectories are nearly invariant. Panel (c) shows nearly identical model-implied income-gap paths across the three specifications. This reflects the reconstruction of wages using the same observed female income shares in each calibration. Panel (d)

shows greater sensitivity in the level of the married-female consumption equivalent: a higher χ produces a larger gap relative to married-male welfare. The declining trajectory associated with wage convergence nevertheless remains common to all three specifications.

The scope for redistribution widens with higher χ, and the Shapley decomposition is qualitatively stable. Panel (e) reports the paternal efficiency threshold $\bar{\eta}_t$ from (18). By 2010, it reaches approximately 0.84 at $\chi = 0.05$, compared with approximately 0.81 in the baseline and roughly 0.79 at $\chi = 0.10$. The higher-time-cost specification can therefore accommodate a larger paternal efficiency shortfall while satisfying the sufficient conditions. A larger care requirement amplifies the effect of redistribution on the wife's effective childcare cost, supporting the husband's participation gain at lower paternal efficiencies. The common trajectory—declining to a trough around 2010 before rising after 2010 as the observed paternal share approaches the boundary of the sufficient-condition region—is present in all three specifications, although at $\chi = 0.05$ the threshold returns to unity by the end of the sample.

Panel (f) shows that the fertility decomposition is also qualitatively stable. The declining wage gap (B, maroon) remains the principal contributor to lower fertility. The offsetting childcare-division contribution (χ-force, green) grows with the assumed time requirement, consistent with a larger effect of reallocating more time-intensive care. Gender-neutral productivity (A, navy) plays a supporting role. Across specifications, wage convergence drives the decline while childcare redistribution partly buffers it.

These exercises preserve the main qualitative findings under alternative childcare time costs. The model continues to fit the targeted demographic moments, the Shapley decomposition retains the same pattern, and the paternal-efficiency boundary declines in all three specifications. The level of that boundary remains sensitive to χ, underscoring the value of reporting the quantitative range alongside the common direction of change.

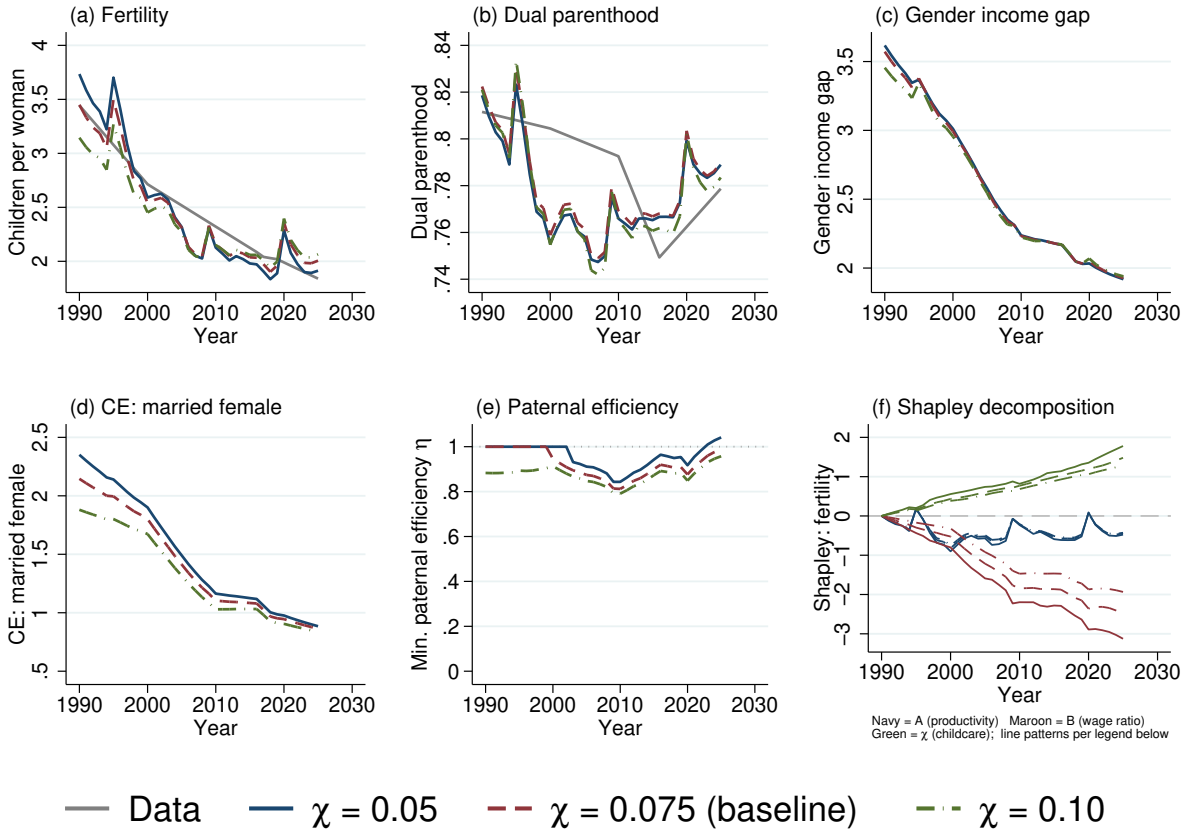


Figure A.6. Robustness of the Calibration to Alternative Values of χ

Notes: Each panel compares the calibrated model under $\chi = 0.05$ (solid navy), $\chi = 0.075$ (dashed maroon, baseline), and $\chi = 0.10$ (dash-dot green), using the parameters in Table A.7. Panels (a) and (b) also plot the data (solid gray). For each χ, wages are re-recovered and the four parameters $(\beta, \rho, \lambda, \theta)$ are re-estimated to match fertility and dual parenthood. Panel (c): gender income gap Γ^y. Panel (d): consumption equivalent for married females benchmarked to the married male. Panel (e): paternal efficiency threshold $\bar{\eta}_t$ from (18), approximated by the smallest passing grid value; the dotted line at $\eta = 1$ marks equal efficiency. Panel (f): Shapley-value decomposition of fertility; colours encode the force (A = navy, B = maroon, χ = green) and line patterns encode the χ-scenario as in the shared legend.

H Why the Wage Gap Organizes the Bottleneck

Lemma 3 identifies two channels through which the gender wage gap Γ^w connects fertility, marriage, and income inequality. OECD and non-OECD countries also differ in productivity levels, fertility preferences, and childcare arrangements. This appendix compares how those alternatives enter the same channels. The analytical comparisons concern the model's structural relationships, while the Shapley contributions provide quantitative context from Mexico and may differ in other economies.

Gender-neutral productivity (A). Consider two economies sharing $(\beta, \rho, \chi, \lambda, \theta)$ and Γ^w but differing in gender-neutral productivity $A = w^{\male}$ (with $w^{\female} = A/\Gamma^w$). The F–G ceiling defined in Lemma 3(i),

$$\bar{n}(\Gamma^w) = \frac{1}{\chi}\left(1 - \frac{\Gamma^w}{\bar{\Gamma}}\right),$$

depends on Γ^w, χ, and the equality threshold $\bar{\Gamma}$, but not on A. This is immediate: the accounting identity $\Gamma^y = \Gamma^w/(1 - \chi n)$ is independent of wage levels. The fertility ceiling compatible with a given degree of gender equality is therefore the same at every productivity level, provided the wage gap is the same.

Productivity does affect the F–M coupling. A higher A raises C (since $\rho > 1$ implies that $(w^{\female})^{\rho-1}$ increases with A), which raises K and hence $g = K^{1/r}$. Since $\partial(q-1)/\partial g < 0$ (Appendix D.5), the marital fertility premium $q - 1$ falls. Higher productivity thus weakens the coupling between marriage and fertility, but it operates through a single channel: $A \to g \to q$. By contrast, Γ^w enters the coupling through two reinforcing channels—directly through the Γ^w terms in $q - 1 = (1 - g)\Gamma^w/(1 + g\Gamma^w)$ and indirectly through g—as established in Lemma 3(ii).

At a fixed wage gap, productivity changes the marital fertility premium while leaving the accounting ceiling on fertility unchanged. It can still affect realized income inequality by changing fertility and labor supply. In the Mexico decomposition, however, this contribution is small: A accounts for 3.9 percent of income-gap convergence, compared with 94.4 percent for B (Table 7). These magnitudes illustrate the relative importance of the channels in the calibrated transition.

Fertility preferences (β). The F–G ceiling $\bar{n}(\Gamma^w)$ follows from the accounting identity $\Gamma^y = \Gamma^w/(1 - \chi n)$ and is independent of preferences. Changes in β affect realized fertility and the marital fertility premium, but leave the ceiling itself unchanged. Preferences therefore influence where an equilibrium lies relative to an income-equality target, while the wage gap changes both that target's fertility constraint and the composition channel characterized in Lemma 3.

Gender division of childcare. Allowing both parents to provide care changes the income identity to $\Gamma^y = \Gamma^w \cdot l^{\male}/l^{\female}$ and affects both mechanisms considered here. When fathers share some of the

childcare ($\chi^{♂} > 0$), the F–G relationship changes in two specific ways relative to the baseline.

First, the ceiling on fertility compatible with a given degree of gender equality relaxes. With paternal care, male labor supply falls below unity ($l^{♂} = 1 - \frac{\chi^{♂}}{\eta}\mathcal{M}n^m < 1$), while female labor supply rises relative to the baseline ($l^{♀} = 1 - \chi n + \chi^{♂}\mathcal{M}n^m > 1 - \chi n$). The ratio $l^{♂}/l^{♀}$ is therefore smaller than $1/(1-\chi n)$, and the maximum fertility consistent with $\Gamma^y \leq \bar{\Gamma}$ increases. The magnitude of this shift, however, is modest at observed childcare divisions. Under the calibrated Mexican parameters at 2025—$\chi^{♂} \approx 0.026$, $\mathcal{M} \approx 0.54$, $n^m \approx 2.96$, and $\eta = 1$—the male labor-supply reduction is approximately $\chi^{♂}\mathcal{M}n^m \approx 0.041$, a roughly 4 percent departure from full market time. This is small relative to the baseline female childcare burden $\chi n \approx 0.15$ that drives the main amplification, leaving the F–G ceiling primarily determined by Γ^w.

Second, the F–M coupling is also affected. With shared care, the marital fertility premium $q = n^m/n^s$ depends on the wife's effective childcare cost $\tilde{\chi}^{♀}$ (equation 13), which itself depends on the childcare division. At a fixed transfer, shifting care toward the husband lowers $\tilde{\chi}^{♀}$ and raises n^m when Condition 1 holds, increasing the premium. The equilibrium transfer adjusts in response, however, and the net effect on q depends on the full set of equilibrium derivatives (Appendix E.2). Unlike the two-channel characterization available for Γ^w in Lemma 3(ii), the childcare channel's effect on the F–M coupling does not admit a simple directional summary without specifying the economic environment.

The Mexico decomposition gives these effects a quantitative context. Changes in the childcare division account for -102.6 percent of the fertility change and -362.1 percent of the dual-parenthood change, but only 1.7 percent of income-gap convergence (Table 7). Thus, childcare reallocation materially offsets demographic declines in this calibration, while the direct relative-wage channel accounts for most observed income convergence.

Summary. Table A.8 collects the comparisons.

Candidate	Affects F–G ceiling?	Affects F–M coupling?	Shapley share (Γ^y)
Γ^w (wage gap)	Yes, directly	Yes, two channels	94.4%
A (wage level)	No	Yes, one channel	3.9%
β (fertility preference)	No	Indirectly (through g)	—
$\chi^{♂}$ (childcare division)	Yes	Yes, direction ambiguous	1.7%

Table A.8. Alternative Candidates for Explaining the Differential Bottleneck

Notes: Each row reports whether the candidate variable affects the F–G ceiling (Lemma 3(i)), whether it affects the F–M coupling (Lemma 3(ii)), and its Shapley share of the income-gap change in the Mexico calibration (Table 7). The F–G ceiling for A and β refers to the baseline accounting identity $\Gamma^y = \Gamma^w/(1-\chi n)$; for the childcare division it refers to the extended-model identity $\Gamma^y = \Gamma^w \cdot l^{♂}/l^{♀}$. The first two columns summarize the model's analytical channels. The Shapley shares are computed under parameters calibrated to Mexico (Table 6); their magnitudes may differ for other economies. β does not vary in the Shapley decomposition and is therefore marked "—".

The comparison highlights the wage gap's role in both the fertility–income equality constraint and the marital-fertility composition channel. Productivity levels and fertility preferences affect equilibrium choices without moving the baseline accounting ceiling. The division of childcare affects both mechanisms in the extended model, with equilibrium effects that depend on the transfer and fertility responses. In Mexico, its contribution to income convergence is 1.7 percent, although its demographic contributions are substantial. These comparisons support the wage gap as an organizing variable for the model's mechanisms while keeping the quantitative ranking specific to the calibrated economy.